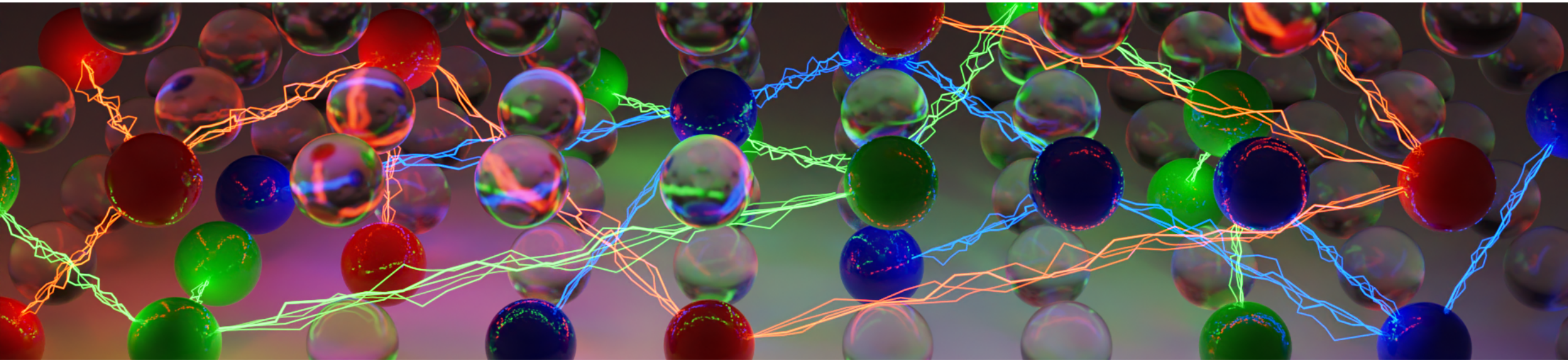
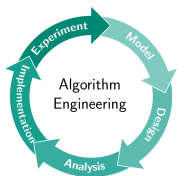




Practical SAT Solving

Lecture 8 – Parallel SAT Solving

Ashlin Iser, Dominik Schreiber | June 16, 2025



SAtRes
Scalable Automated Reasoning

Overview

Recap Lecture 7

- Propagation-based Redundancy notions
- Proof systems: Resolution, Extended Resolution, Blocked Clauses, Implication-based Redundancy
- Autarkies, Conditional Autarkies, and Satisfaction Driven Clause Learning (SDCL)

Today: Parallel SAT Solving

Parallel SAT solving approaches

- Basic search space splitting, Clause sharing, Cube&Conquer, Portfolio solvers

A deep dive into Mallob

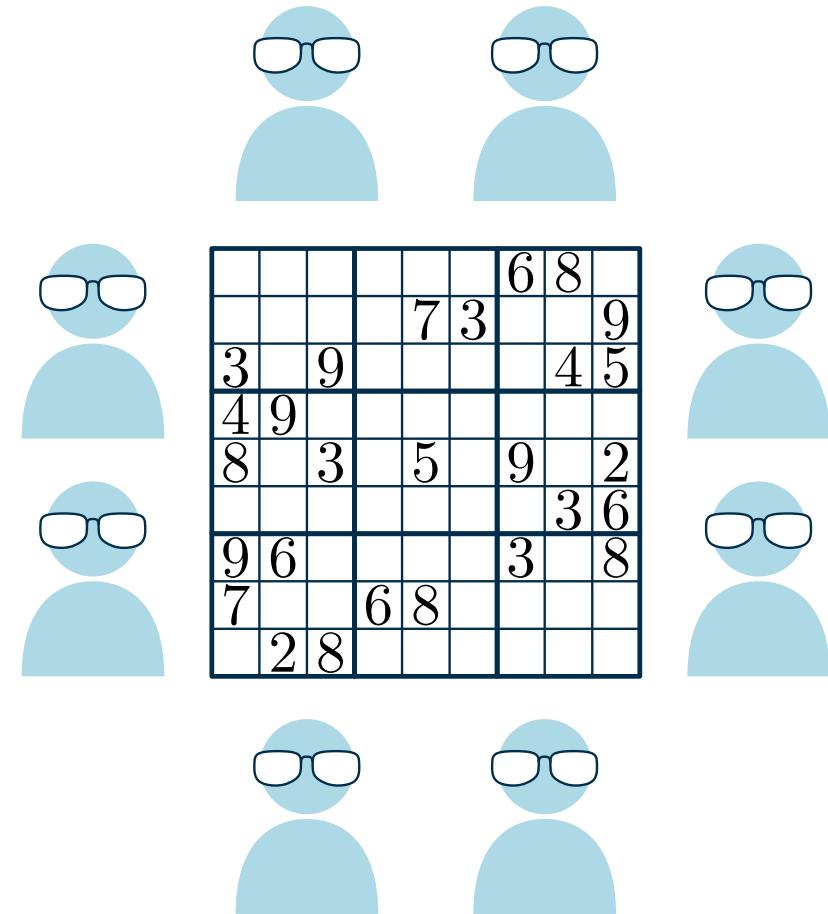
- Overview, Scalable clause sharing, Experiments and insights

Parallel Solving: An analogy

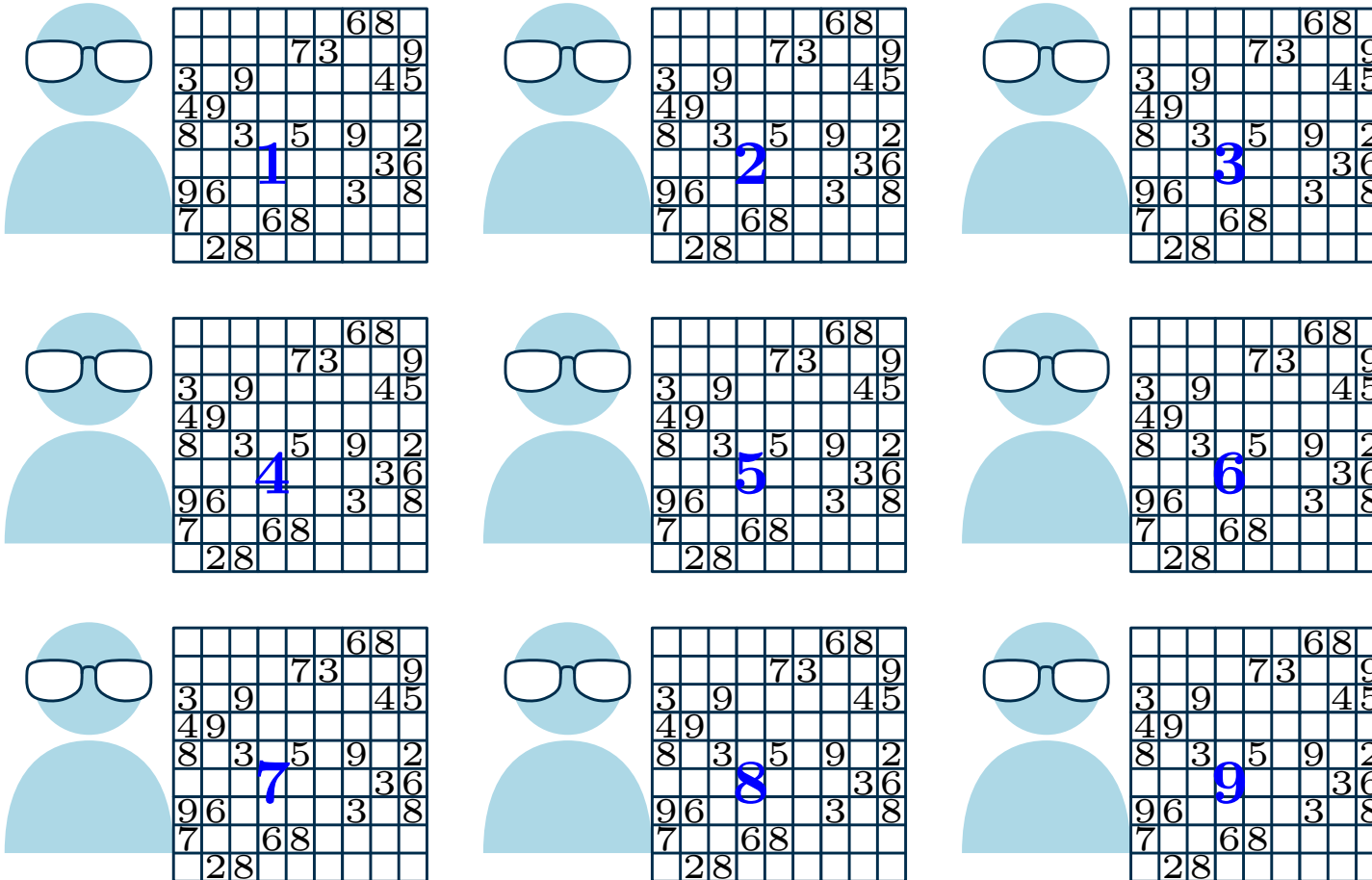
The Assembly of Nerds [23]

- Complex and large logic puzzle
- n puzzle experts at your disposition
 - **anti-social**: work best if left undisturbed

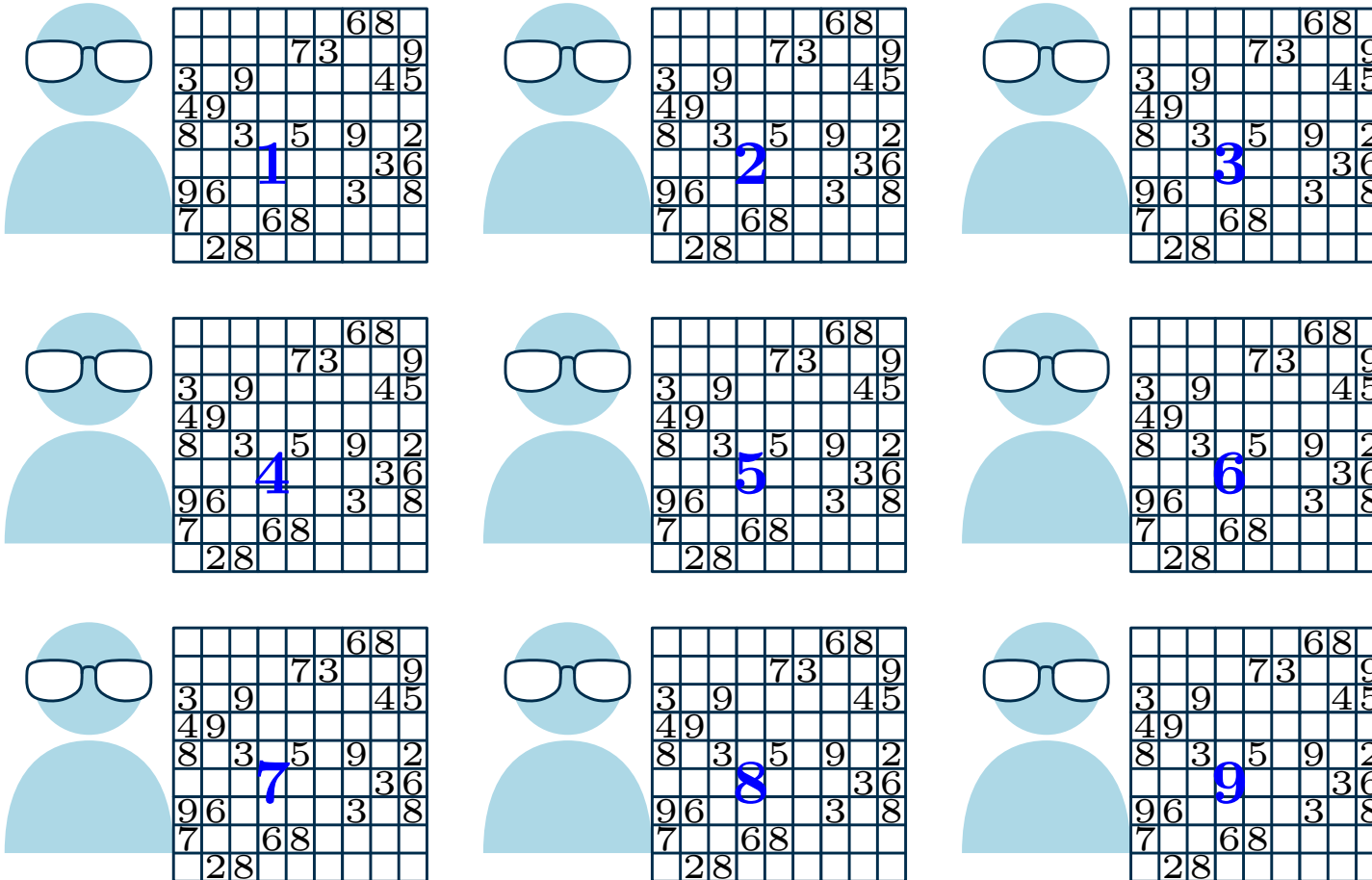
How do we employ and “orchestrate” our experts?



Approach I: Search Space Partitioning



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- Partition search space at some decisions
⇒ Independent subproblems

Explicit Partitioning

Böhm & Speckenmeyer (1994-1996) [5]: 1st Parallel DPLL Implementation

Explicit Load Balancing

- Completely distributed (no leader / worker roles)
- A list of **partial assignments** is generated
- Each process receives the entire formula and **a few partial assignments**
- Each process can be worker or balancer:
 - **Worker**: solve or split the formula, use the partial assignments
 - **Balancer**: estimate workload, communicate, stop
- Switch to balancer whenever worker is finished

Explicit Partitioning

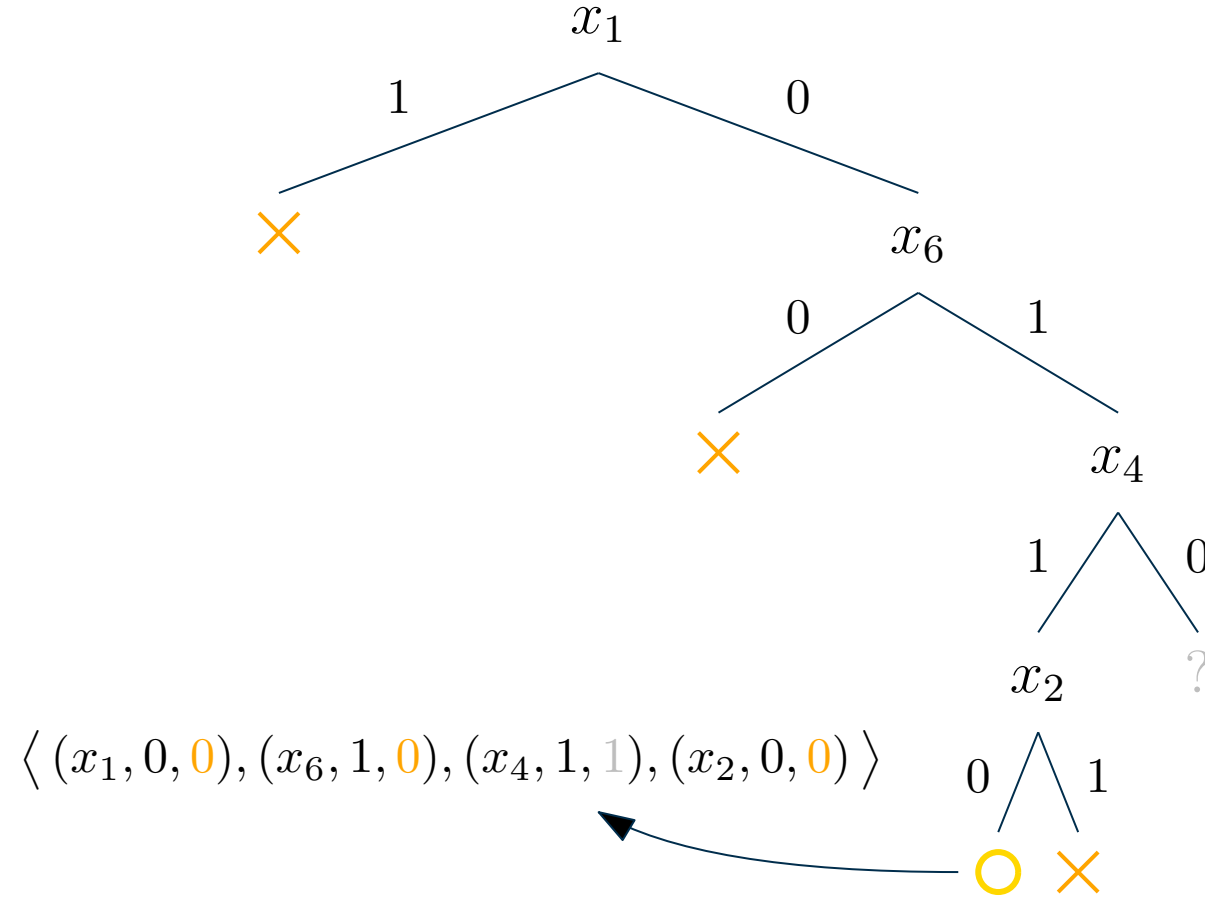
“**PSATO**: a Distributed Propositional Prover and its Application to Quasigroup Problems”, Zhang et al. (1996) [26]

Centralized leader-worker architecture

- Communication only between leader and workers
- Leader assigns partial assignments using **Guiding Path**
 - Each node in the search tree is **open** or **closed**
 - closed = branch is **explored** / **proven unsat**
 - Leader splits open nodes and assigns job to workers
- Workers return **Guiding Path** when terminated by leader
- Modern features of **fault tolerance**, **preemption** of solving tasks

Explicit Partitioning

Guiding Path: List of triples (variable, branch, open)



Explicit Partitioning Solvers

SATZ (Jurkowiak et al., 2001) [15]: *Work stealing* for workload balancing

- An idle worker requests work from the leader
- The leader splits the work of the most loaded worker
- The idle worker and most loaded worker get the parts

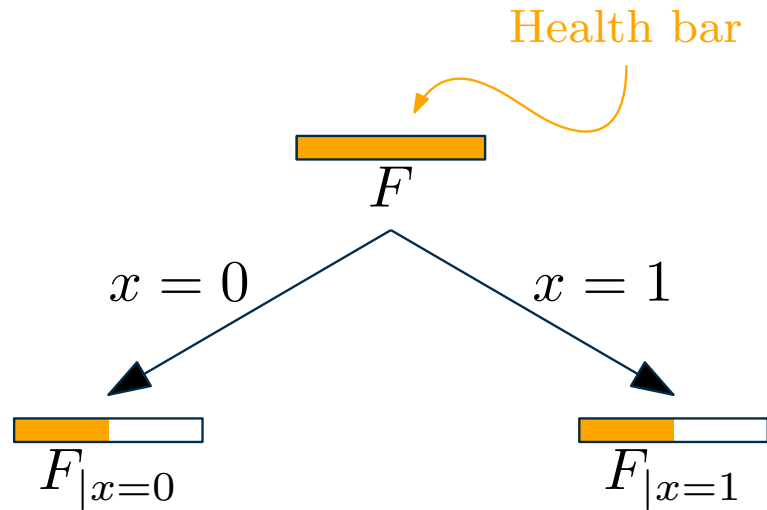
PaSAT (Blochinger et al., 2001-2003) [4]

- First parallel CDCL with clause sharing
- Similar to PSATO/SATZ: leader/worker, guiding path, work stealing

ySAT (Feldman et al., 2004) [8]

- First shared-memory parallel solver
- Multi-core processors started to be popular
- uses same techniques as the previous solvers (guiding path etc.)

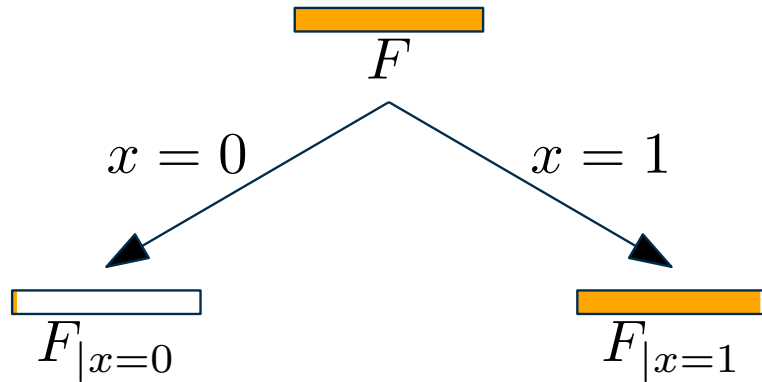
Problems with Partitioning [25]



What we want: **Even splits**

- Split yields sub-formulas of **similar difficulty**
- **Balanced** partitioning of work
- Few or no dynamic (re-)balancing needed

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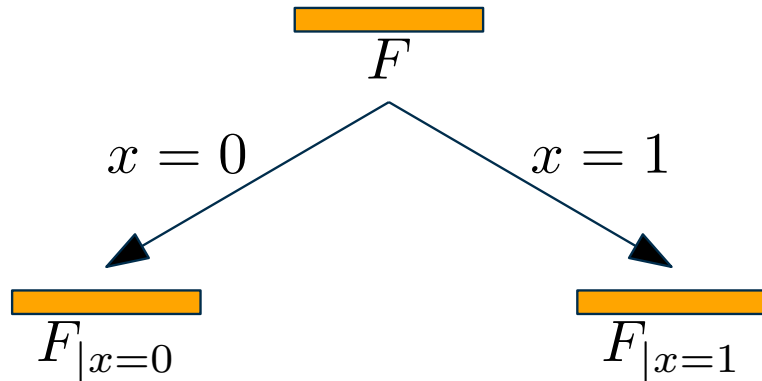
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Uneven splits

- One subformula is **trivial**, the other is **just as hard as F**
- **Ping-pong effect** for workers processing trivial formulae, communication / synchronization dominates run time

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Bogus splits

- Both $F|_{x=0}$ and $F|_{x=1}$ are **just as hard as F**
- **Divide&Conquer** becomes **Multiply&Surrender!**

Cube and Conquer

The Cube&Conquer paradigm (Heule & Biere, 2011) [14]

Generate a large amount (millions) of partial assignments (“**cubes**”) and randomly assign them to workers.

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- Partial assignments are generated using a look-ahead solver (breadth-first search up to a limited depth)
- Best performance mostly with problem-specific decision heuristics

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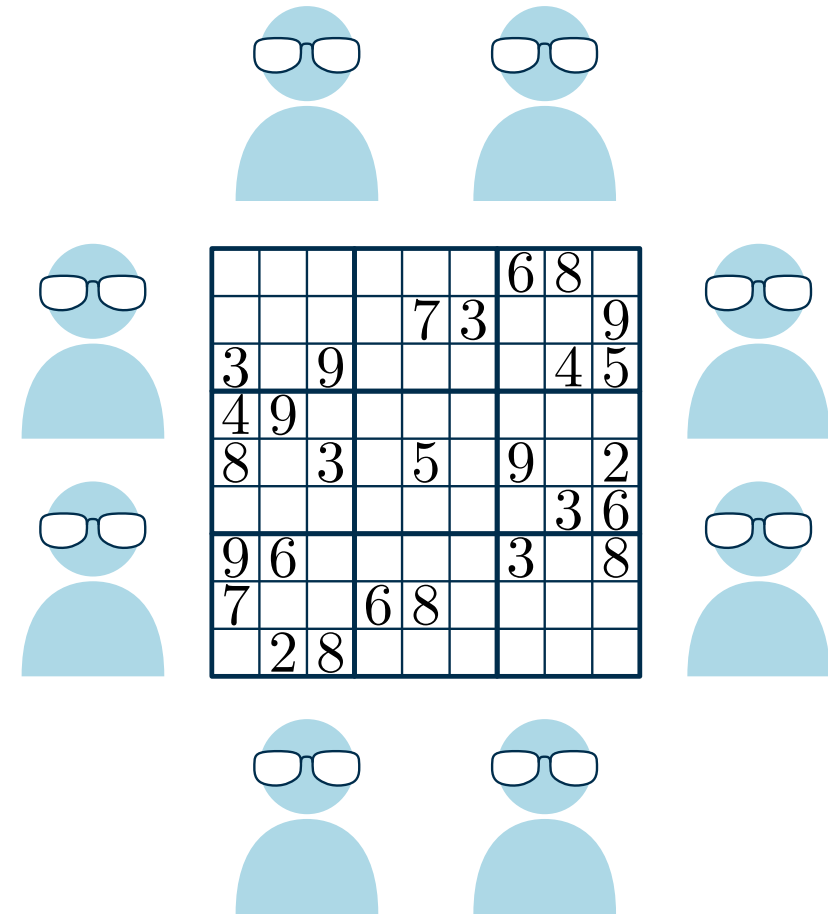
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- Partial assignments are generated using a look-ahead solver (breadth-first search up to a limited depth)
- Best performance mostly with problem-specific decision heuristics
- State-of-the-art for hard combinatorial problems
 - Used to solve the “Pythagorean Triples” problem (~200TB proof) [13]
 - ... or more recently “Schur Number 5” (~2PB proof) [12]
- Examples: March (Heule) + iLingeling (Biere) introduced in 2011; Treengeling (Biere) [2]

Parallel Portfolios: An analogy

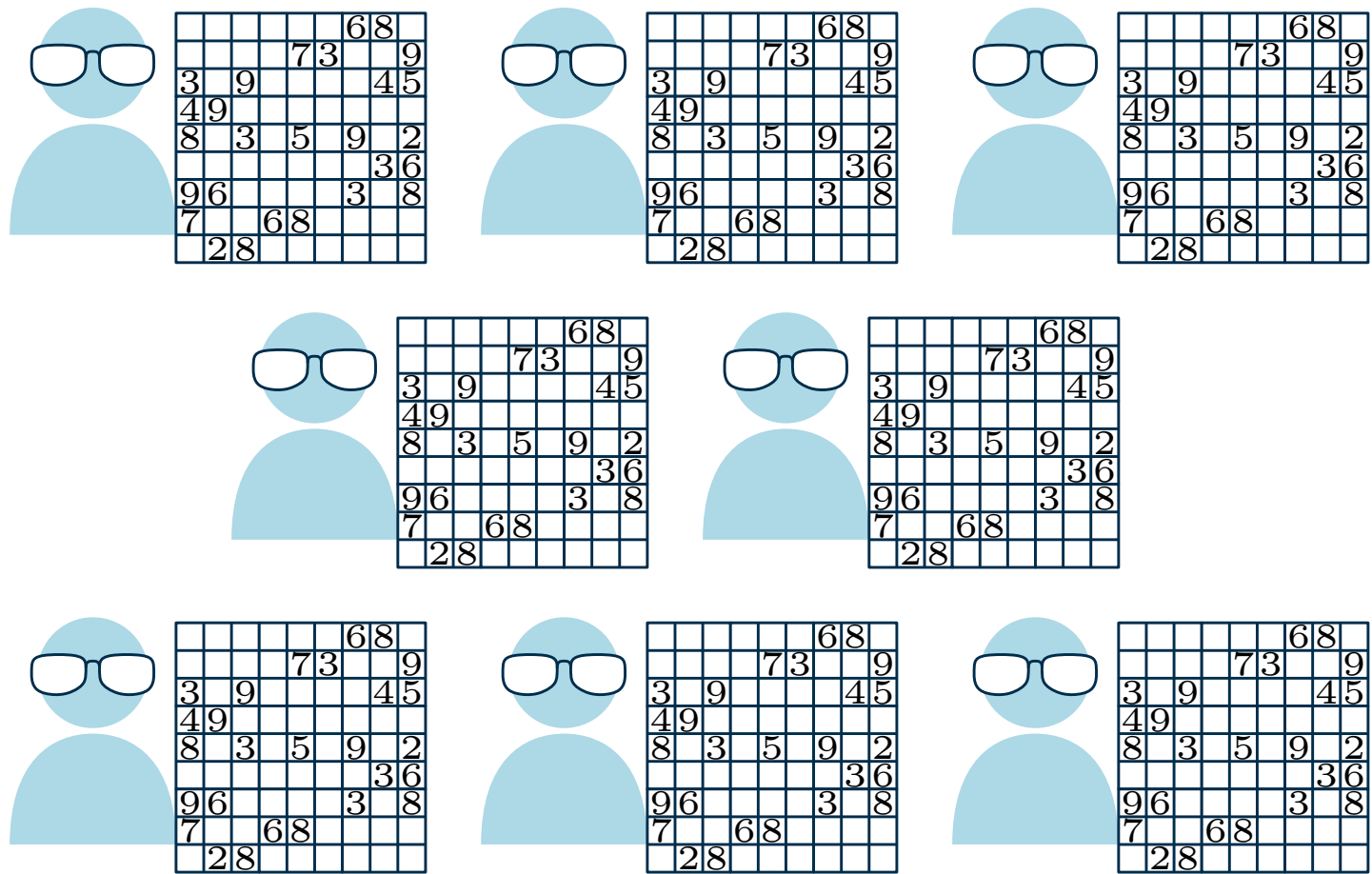
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- Complex and large logic puzzle
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 - individual mindsets, approaches, strengths & weaknesses
 - **anti-social**: work best if left undisturbed

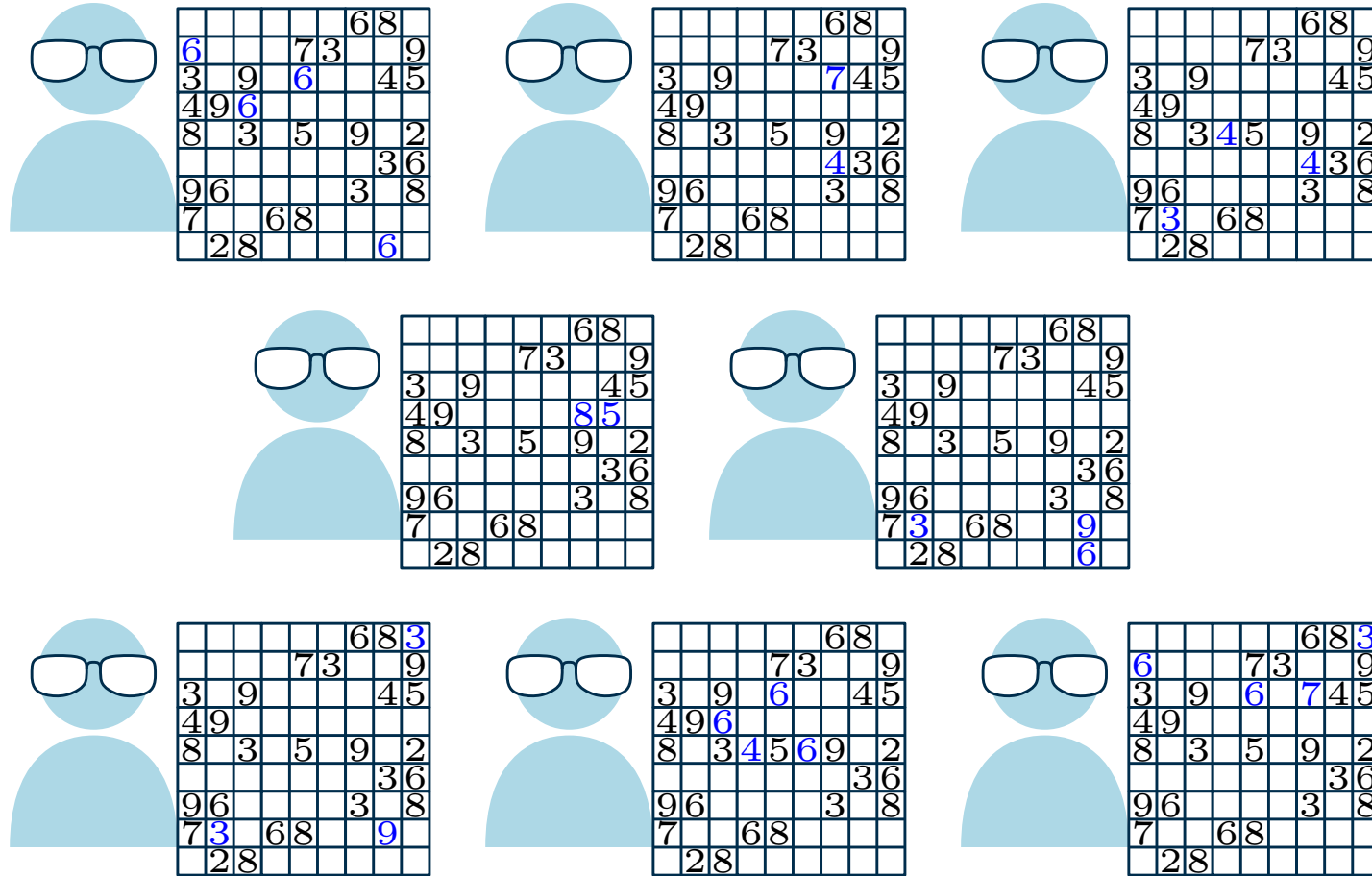
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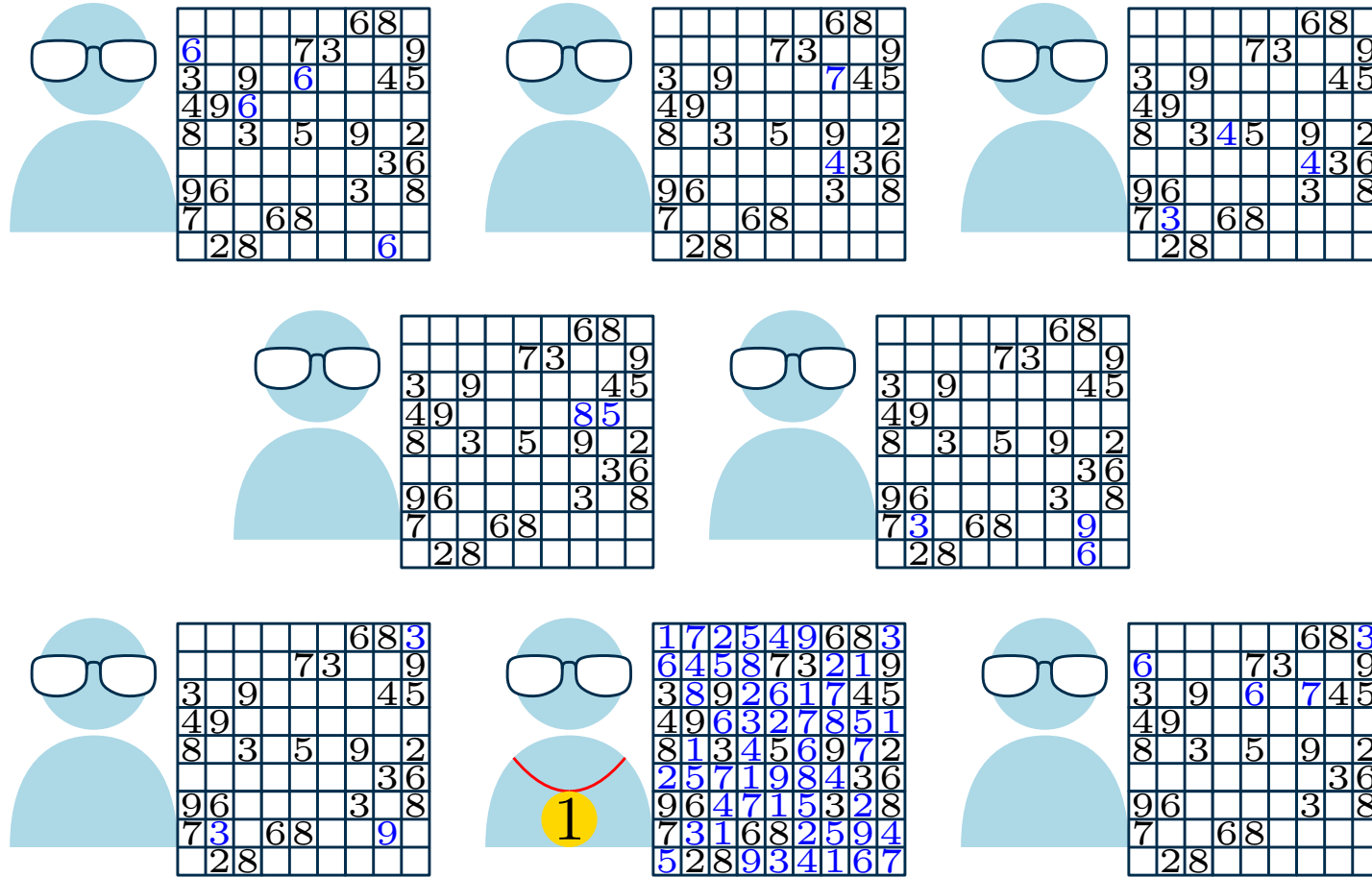
Approach II: Pure Portfolio



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Approach II: Pure Portfolio



Pure Portfolio: Oracle view vs. Speedup view

Virtual Best Solver (VBS) / Oracle

Consider n algorithms $A_1, \dots, A_n$ where for each input x , algorithm A_i has run time $T_{A_i}(x)$.
The **Virtual Best Solver (VBS)** for $A_1, \dots, A_n$ has run time $T^*(x) = \min\{T_{A_1}(x), \dots, T_{A_n}(x)\}$.

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Parallel speedup [20]

Given parallel algorithm P and input x , the **speedup** of P is defined as $s_P(x) = T_Q(x)/T_P(x)$ where Q is the **best available sequential algorithm**.

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Pessimist: A pure portfolio never achieves actual speedups!

- There is always a sequential algorithm performing at least as well
- Consequence: Not resource efficient, not scalable

Pure SAT Portfolios

ppfolio: Winner of Parallel Track in the 2011 SAT Competition [18]

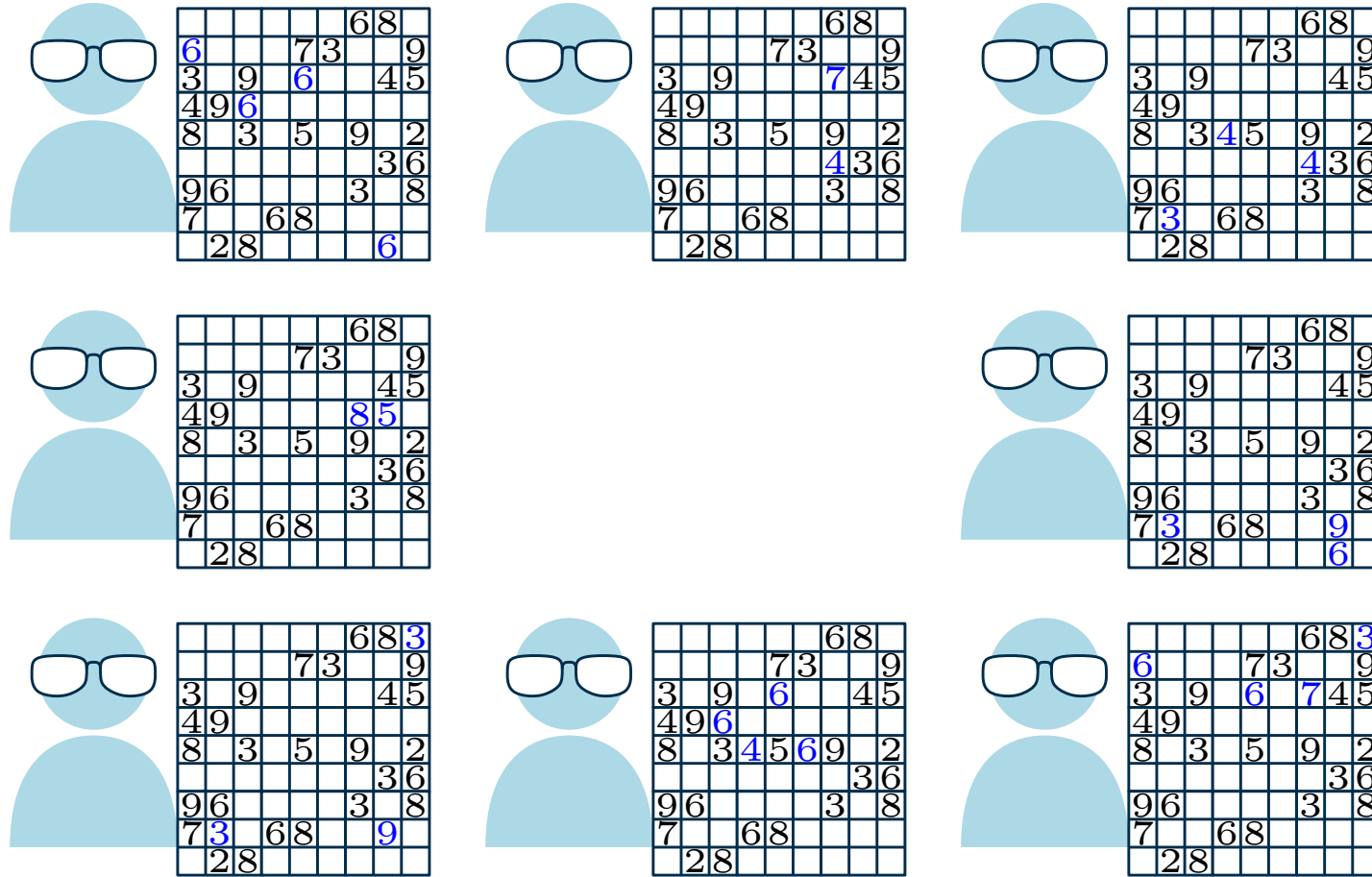
- Just a bash script combining the best sequential solvers from 2010:
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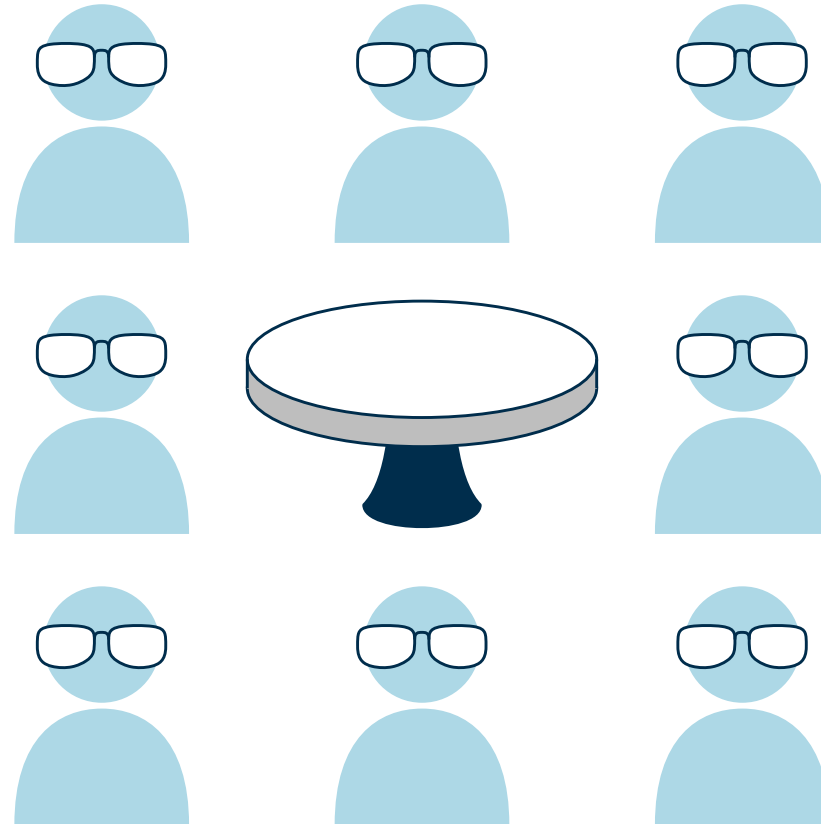
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 - “*by definition the best solver on Earth*”
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- Rationale: Different solvers are designed differently, excel on different instances
 - hope of orthogonal search behavior
- Pure portfolios no longer permitted in SAT Competitions

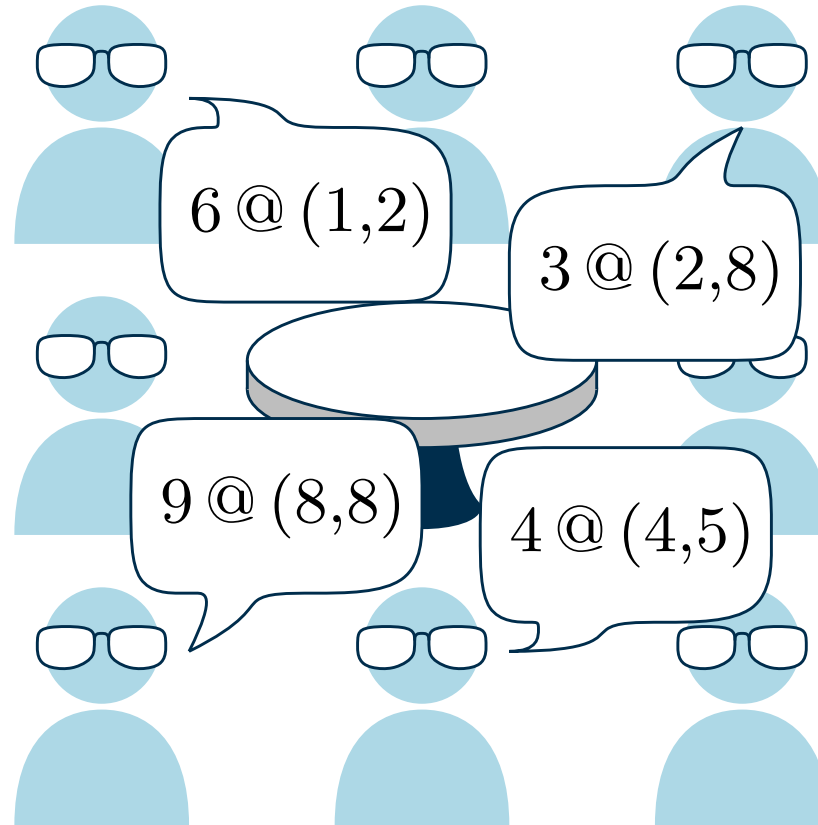
Approach II+: Cooperative Portfolio



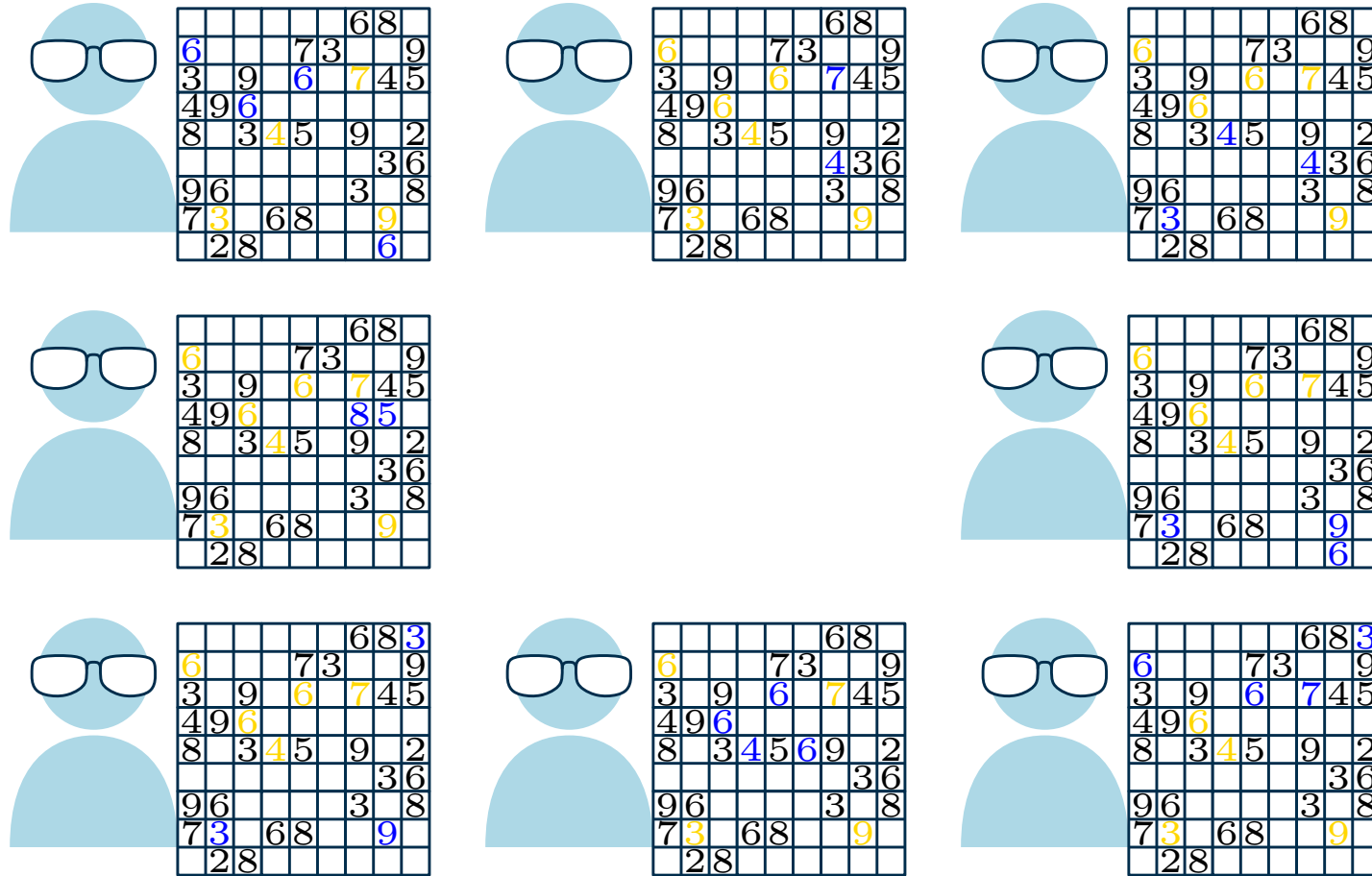
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Approach II+: Cooperative Portfolio



Cooperative Portfolio

Assembly of Nerds, enhanced

- The experts periodically gather for brief standup meetings
- Via some protocol, the experts exchange the most valuable insights gained since the last meeting
- Solving continues — each expert may use the shared insights at their own discretion

Equivalent to “insights” in SAT solving:

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Equivalent to “insights” in SAT solving: **learnt (conflict) clauses**

- Explored branch of search space — **safe to prune**
- Potential step for deriving **unsatisfiability**
- Result: **Parallel search space pruning procedure**

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Combination of portfolio idea + clause exchanges: **Clause sharing portfolio solvers**

Clause Sharing Portfolios: Design Space

Portfolio considerations

- Which sequential solvers to employ?
- How to diversify solvers?
 - different search algorithms, selection heuristics, restart intervals, ...
 - different random seeds, initial phases, input permutations, ...

```
void Cadical::diversify(int seed) {  
    solver->set(name: "seed", val: seed);  
    switch (getDiversificationIndex() % getNumOriginalDiversifications()) {  
    case 0: okay = solver->set(name: "phase", val: 0); break;  
    case 1: okay = solver->configure("sat"); break;  
    case 2: okay = solver->set(name: "elim", val: 0); break;  
    case 3: okay = solver->configure("unsat"); break;  
    case 4: okay = solver->set(name: "condition", val: 1); break;  
    case 5: okay = solver->set(name: "walk", val: 0); break;  
    case 6: okay = solver->set(name: "restartint", val: 100); break;  
    case 7: okay = solver->set(name: "cover", val: 1); break;  
    case 8: okay = solver->set(name: "shuffle", val: 1) && solver->set(name: "  
    case 9: okay = solver->set(name: "inprocessing", val: 0); break;
```

Clause Sharing Portfolios: Design Space

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Clause exchange considerations

- How often to share? (immediate/eager? delayed/lazy? periodic?)
- How many clauses to share? (fixed volume? fixed quality criteria?)
- Which clauses to share? (shortest? lowest LBD?)
- How to implement sharing? (all-to-all? leader-worker? some communication graph?)

Early Clause Sharing Portfolios

ManySAT (Hamadi, Jabbour, and Sais 2009) [11]

- Hand-crafted diversification of **four solver configurations**
 - Restart policy, variable + polarity selection heuristic, ...
- Eager exchange of clauses of length ≤ 8 via **lockless queues**

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Plingeling (Biere 2010) [3]

- Portfolio over **Lingeling configurations** (shared-memory parallelism)
- Lazy exchange of information over “boss thread”
 - 2010: **Unit clauses** only
 - 2011: Unit clauses + **equivalences**
 - Since 2013: Unit clauses + equivalences + **clauses of length ≤ 40 , LBD ≤ 8**
- **Best parallel solver** for many years

Massively parallel hardware?

Distributed computing, High-Performance Computing (HPC)

In distributed computing, several machines (with **no shared main memory**) run together. On each machine we run a number of processes, each of which runs on a number of cores. Processes commonly communicate by exchanging messages.



SuperMUC-NG: 6 336 nodes $\times$ 48 cores

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Large distributed systems (hundreds to thousands of cores) impose new requirements, challenges:

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- Diminishing returns due to **exhausted diversification** of solvers

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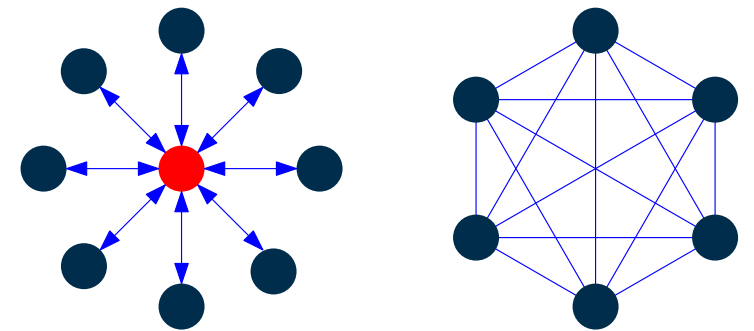
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Large distributed systems (hundreds to thousands of cores) impose new requirements, challenges:

- No shared memory — **communication protocols** required
- Diminishing returns due to **exhausted diversification** of solvers
- Some exchange schemes are **conceptually not scalable** [7]
 - “**Star graph**”: Master process collects, serves all exported clauses
 - **Naïve (quadratic) all-to-all exchange** of clauses
- HPC schedulers, administrators, committees are used to **regular, easily parallelizable code** with **near-linear scaling**



Massively parallel SAT portfolio

HordeSat (Balyo, Sanders, Sinz 2015) [1]

- Decentralization: No single leader node / process
- Two-level (“hybrid”) parallelization
 - One or several processes on each machine
 - Multiple solver threads (+ communication thread) on each process

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- Decentralization: No single leader node / process
- Two-level (“hybrid”) parallelization
 - One or several processes on each machine
 - Multiple solver threads (+ communication thread) on each process
- Diversification options:
 - Native diversification (set of hand-crafted solver configurations)
 - Modifying some initial variable phases
 - Random seeds
- Periodic all-to-all clause exchange

Clause Exchange in HordeSat

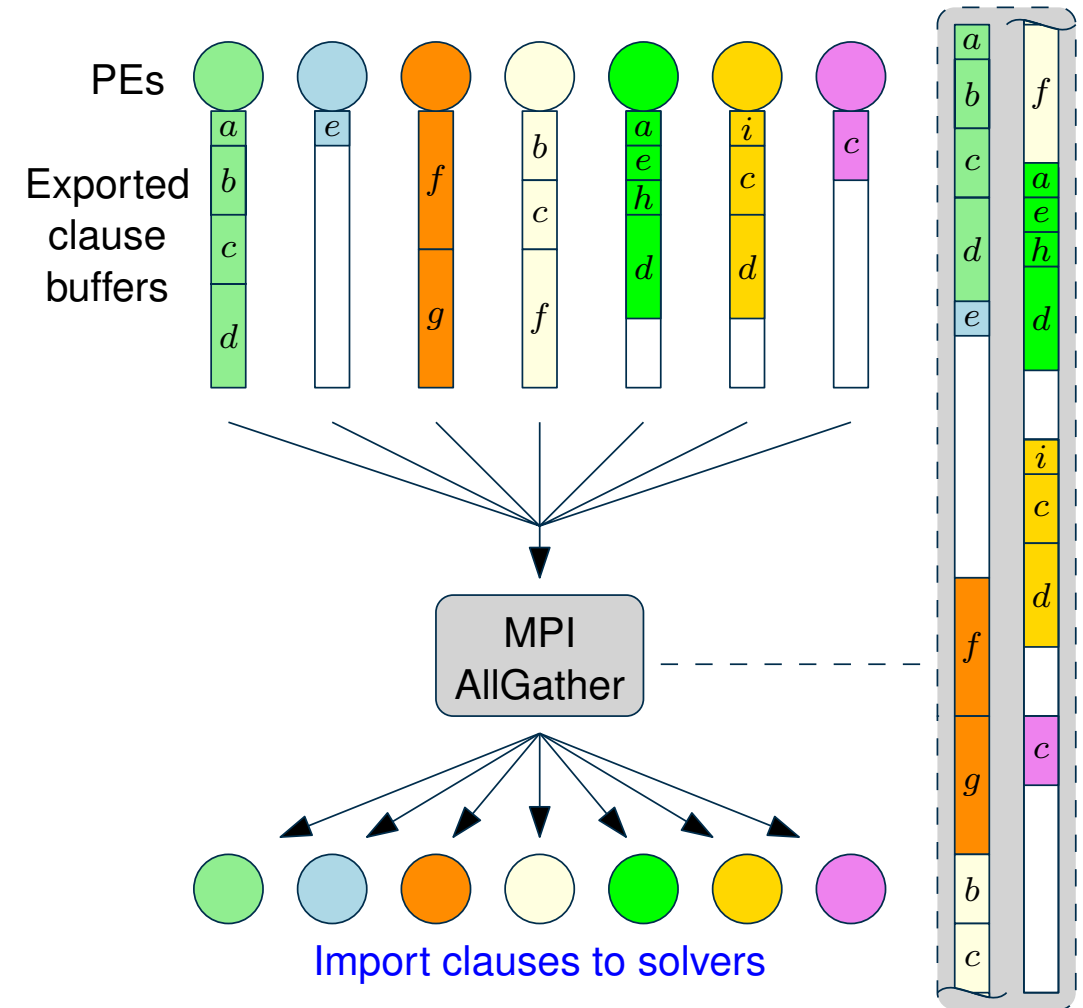
HordeSat's sharing logic

- Only clauses with $LBD \leq 2$ are considered for sharing
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 - Each process' **best clauses** are shared with everyone
 - Limited to 1500 clause literals per process
- ⇒ Concatenation of p produced clause buffers
- Approximate, post-hoc **filtering** of clauses



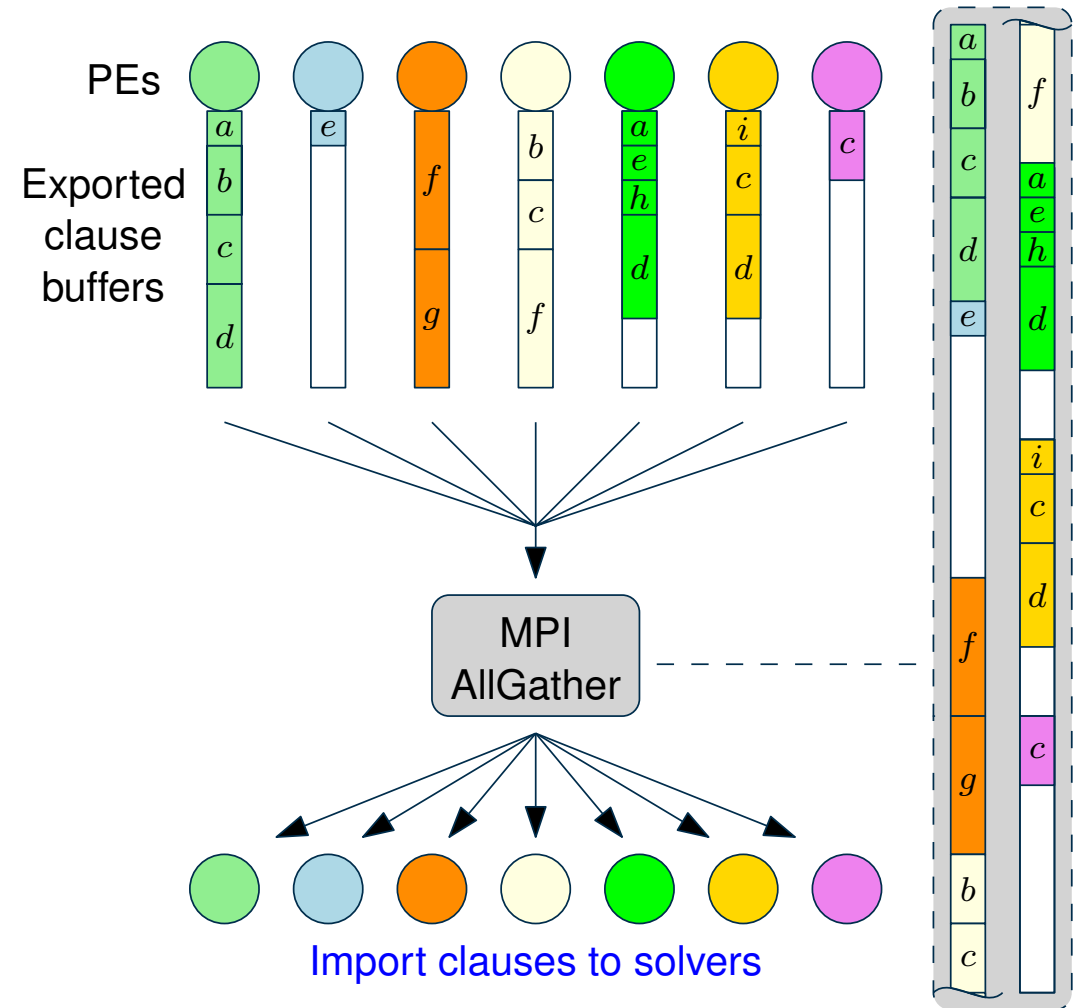
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Issues:

- Many ("high" LBD) clauses are not shared but **discarded**
- **"Holes"** in buffer carrying no information
- **Duplicate** clauses
- Buffer grows **proportionally** with # proc.
 - ⇒ **Bottleneck** w.r.t. communication, local work



HordeSat: Results

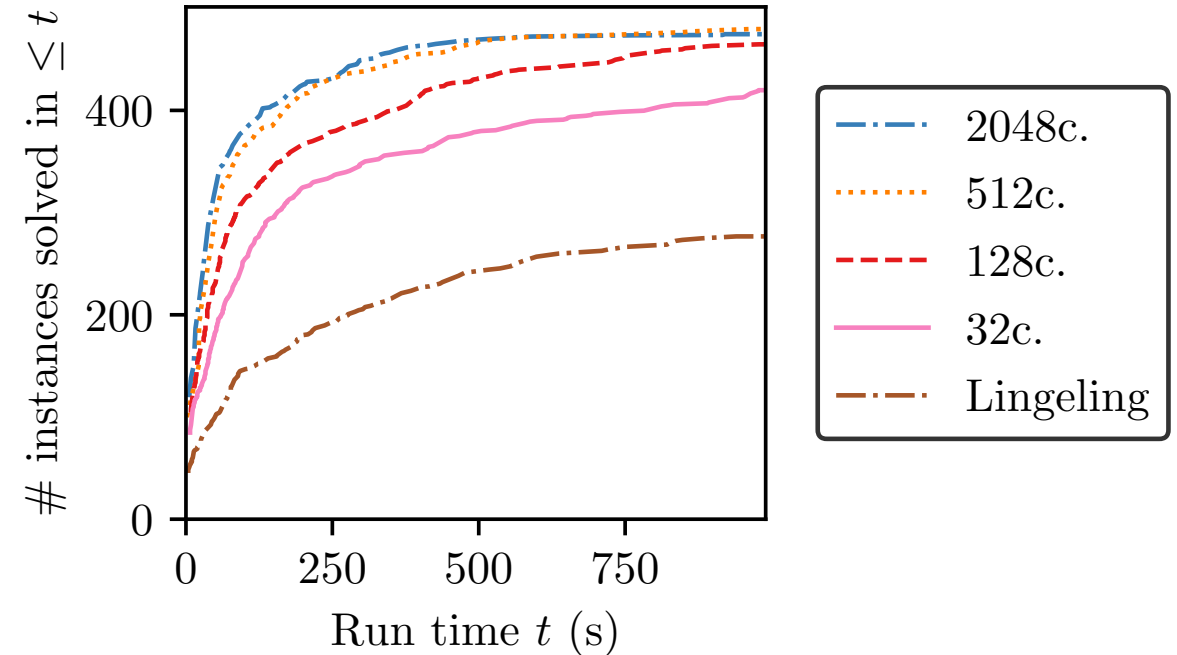
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 - SAT: “NP luck” – some solver got lucky
 - UNSAT: distributed memory accommodates more clauses than any sequential solver
 - General: sequential schedule of parallel algorithm may outperform sequential algorithm!
- Median speedup: 3 at 16 cores, 11.5 at 512 cores
 - Efficiency: $11.5/512 \approx 2.2\%$
 - Deploying HordeSat is often not worth it
- No improvement beyond ≈ 500 cores



Data extracted from HordeSat paper [1]

From HordeSat to MallobSat

Research Question

How can we improve performance, (resource-)efficiency, and average response times of SAT solving in modern distributed environments?

From HordeSat to MallobSat

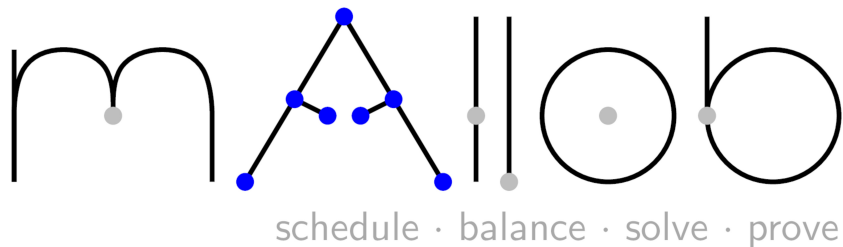
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Result: Mallob

Mallob is a platform for SAT solving (*and other NP-hard problems*) with:

- multi-user, on-demand, malleable scheduling and solving of many problems at once [19]
- distributed SAT engine **MallobSat**: the HordeSat paradigm re-engineered and made efficient [24]
- state-of-the-art SAT performance from dozens to thousands of cores [23]

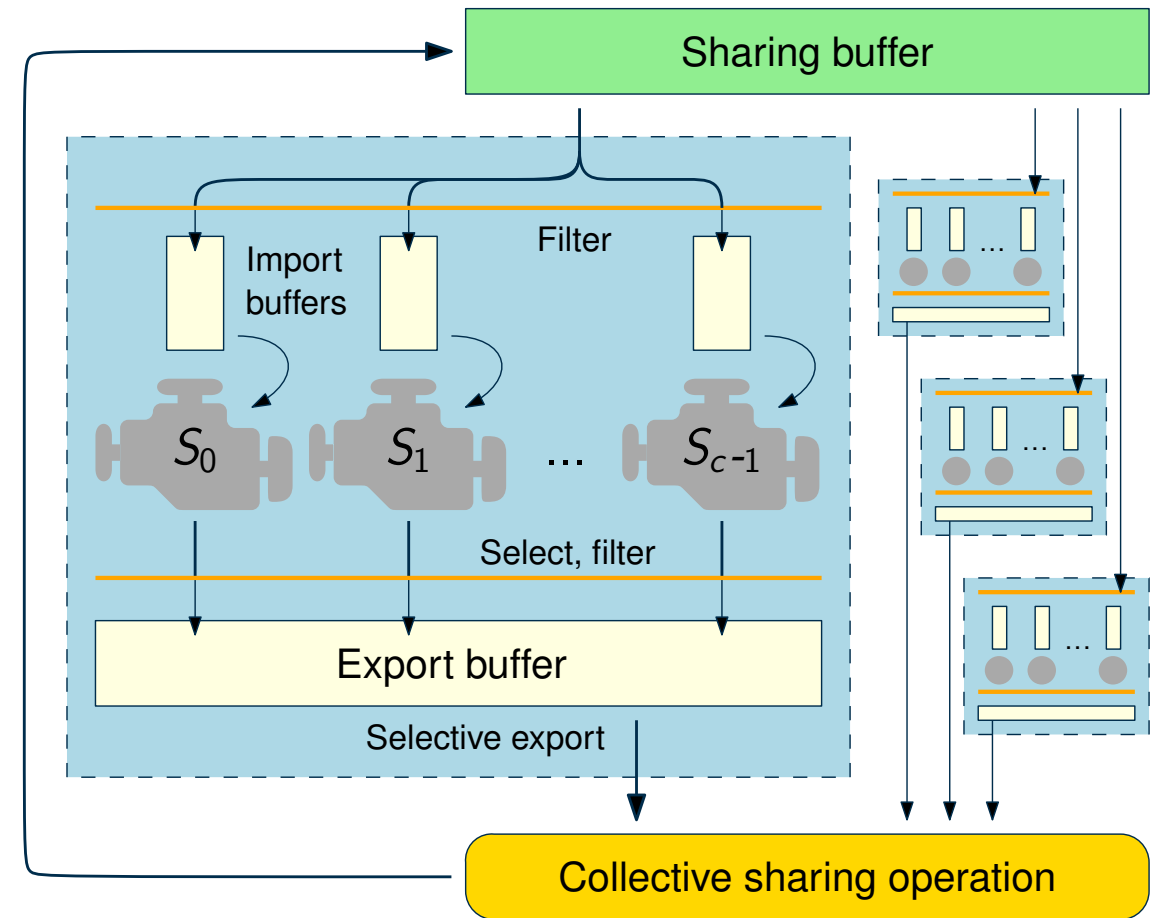


<https://satres.kikit.kit.edu/research/mallob>

Design Decisions of MallobSat [23]

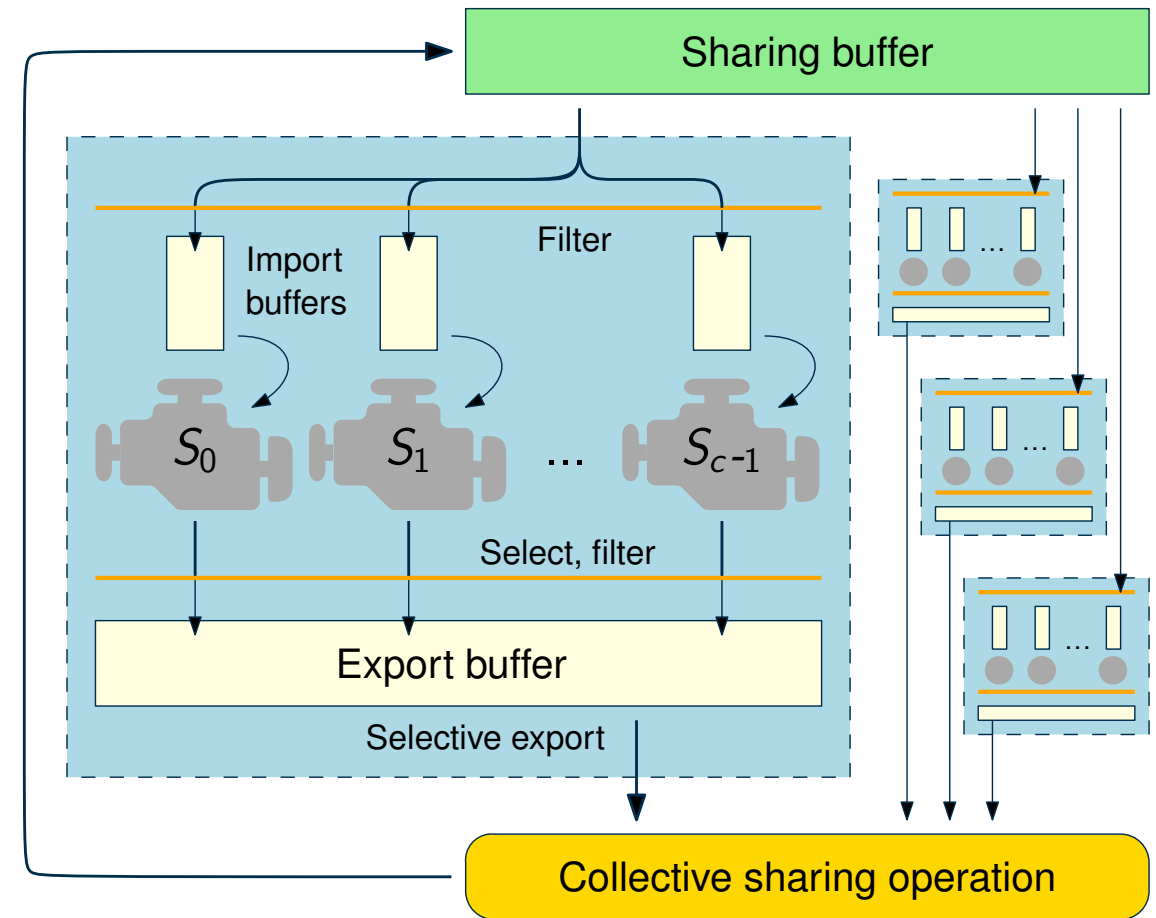
■ **Fundament:** HORDESAT

- Two-level hybrid parallelization
- Periodic all-to-all clause sharing



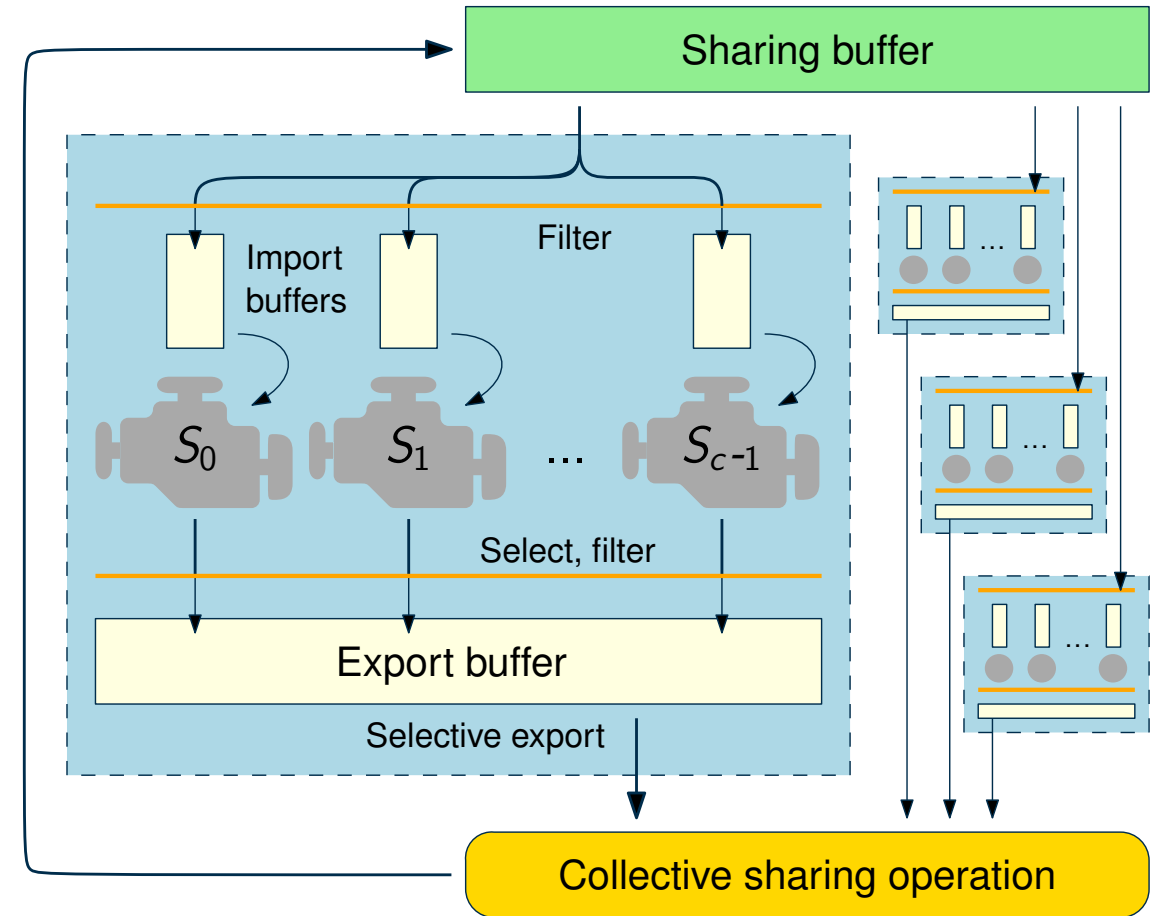
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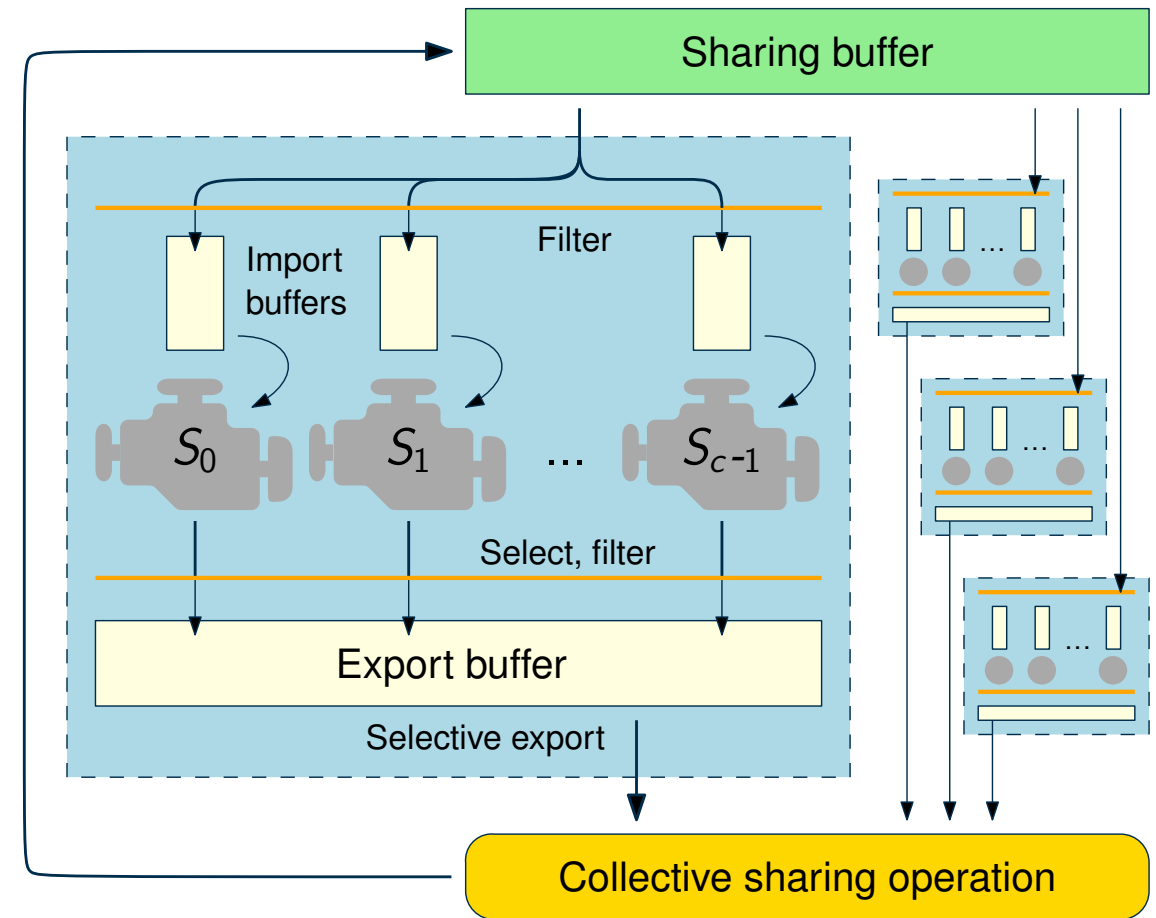
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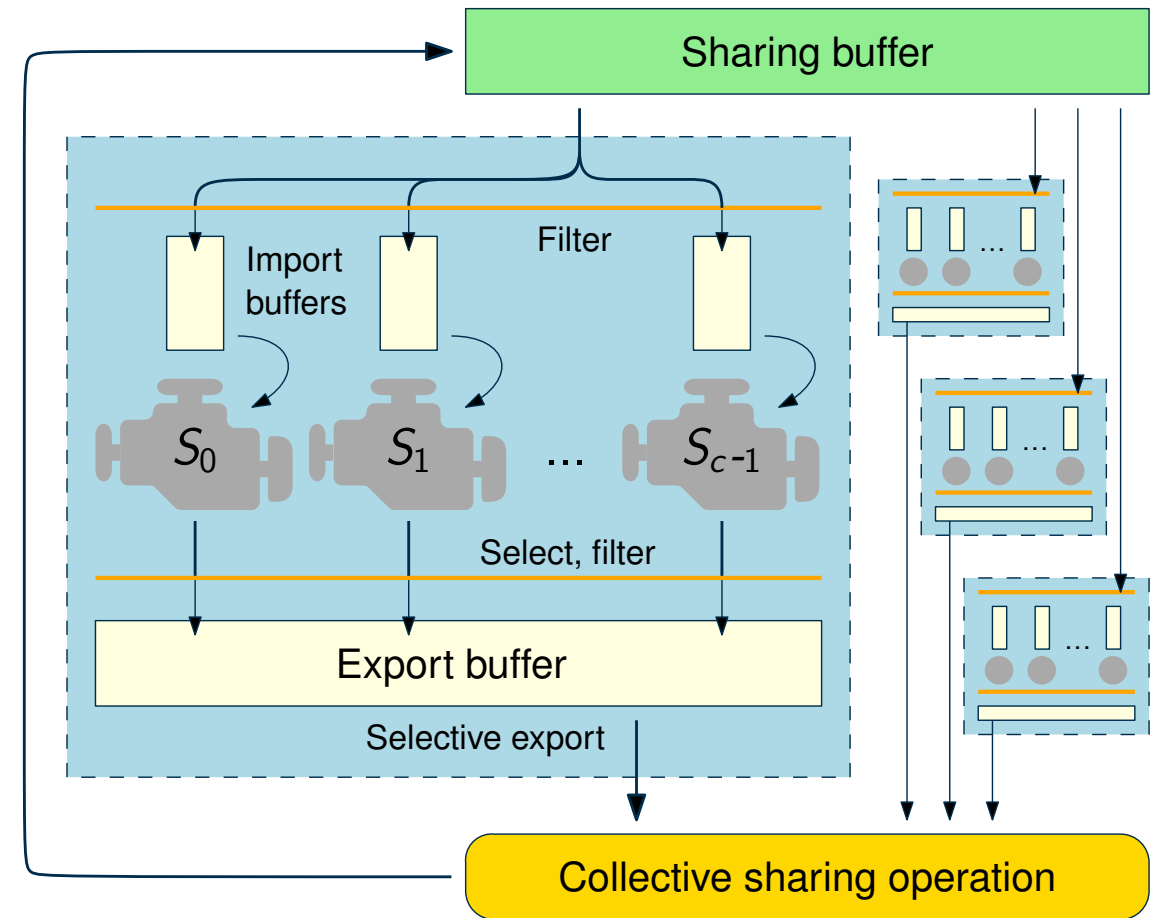
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- Minimize clause turnaround times
- Support **fluctuating workers** (*malleability*)

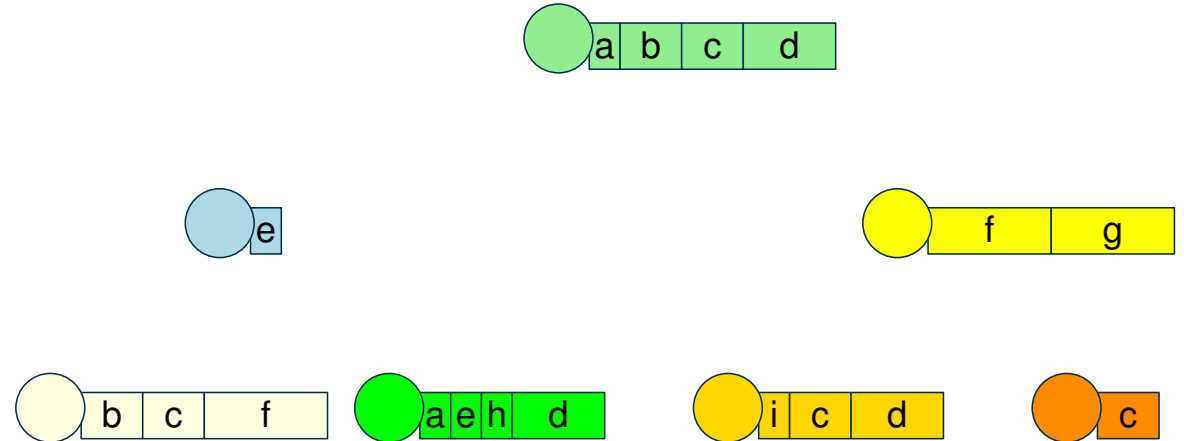


Clause Exchange in Mallob [23]

Custom collective operation

- Aggregate information along binary tree of processors
- Detect duplicates during merge
- Result is of compact shape
- Sublinear buffer size growth:
Discard longest clauses as necessary

1.

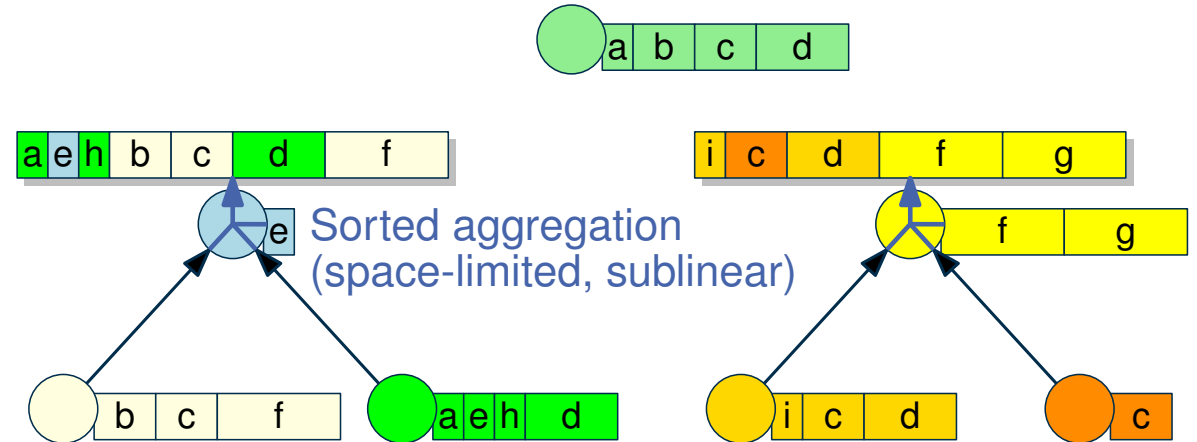


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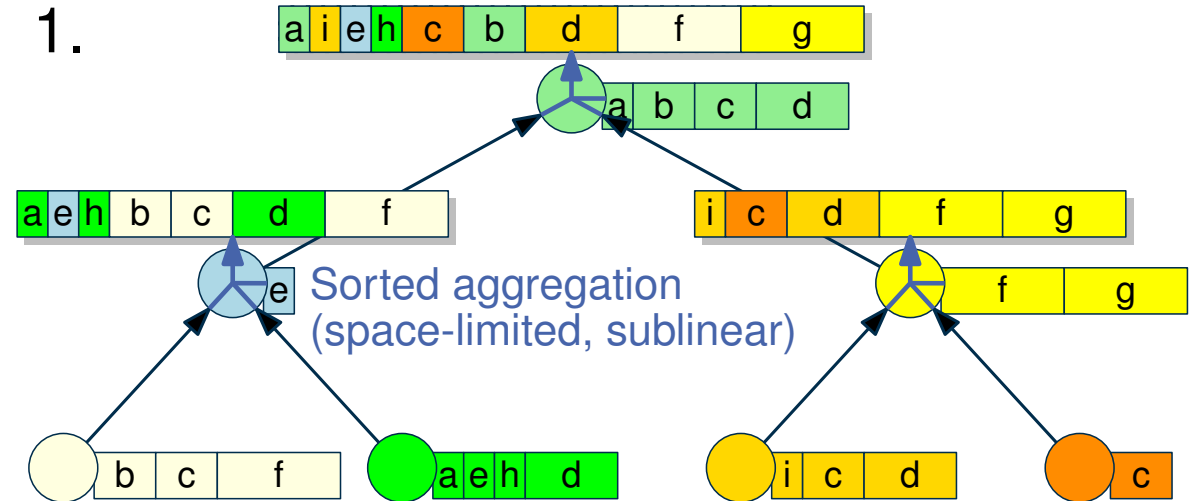
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Clause Exchange in Mallob [23]

Custom collective operation

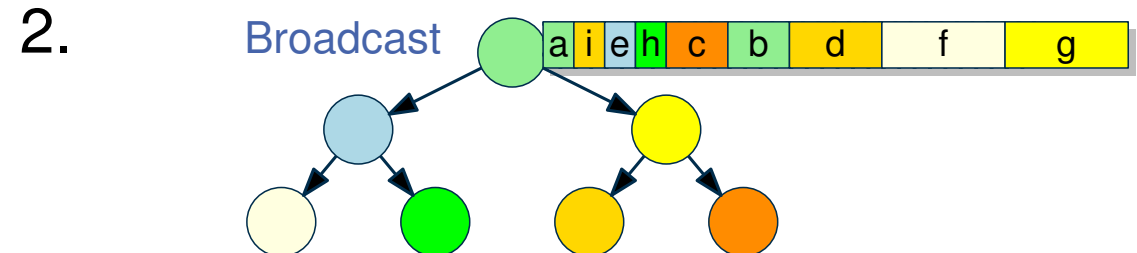
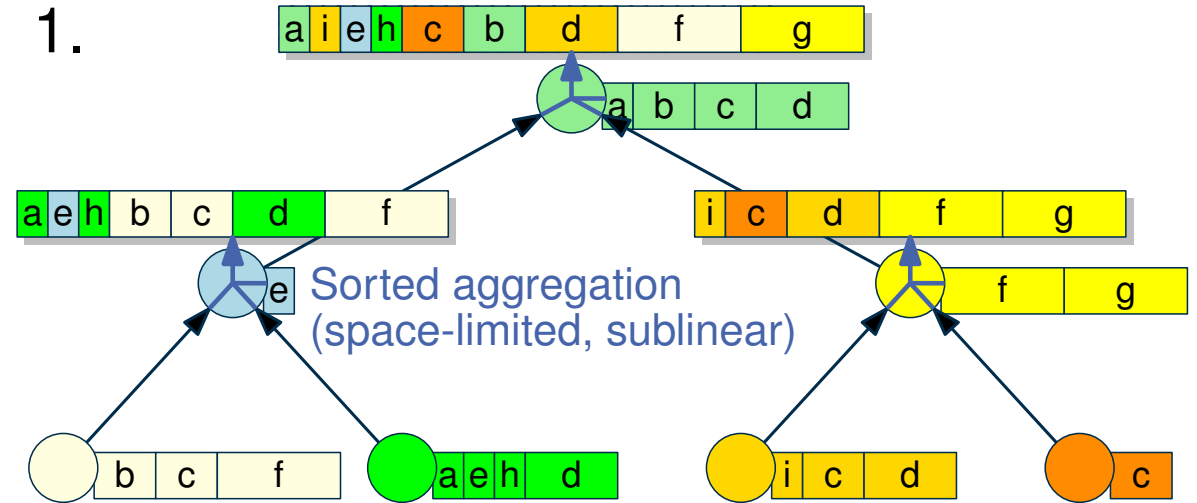
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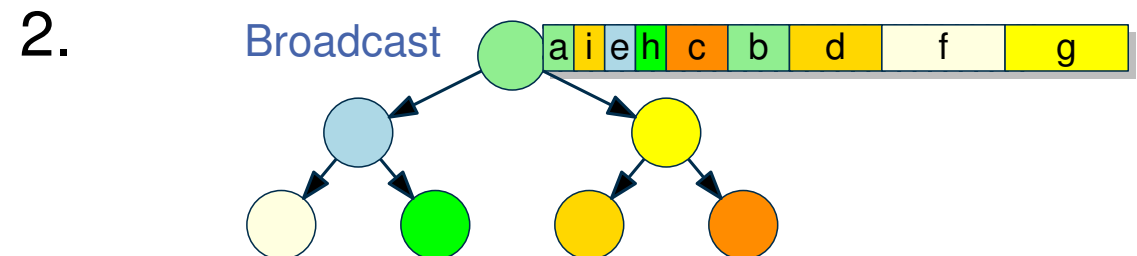
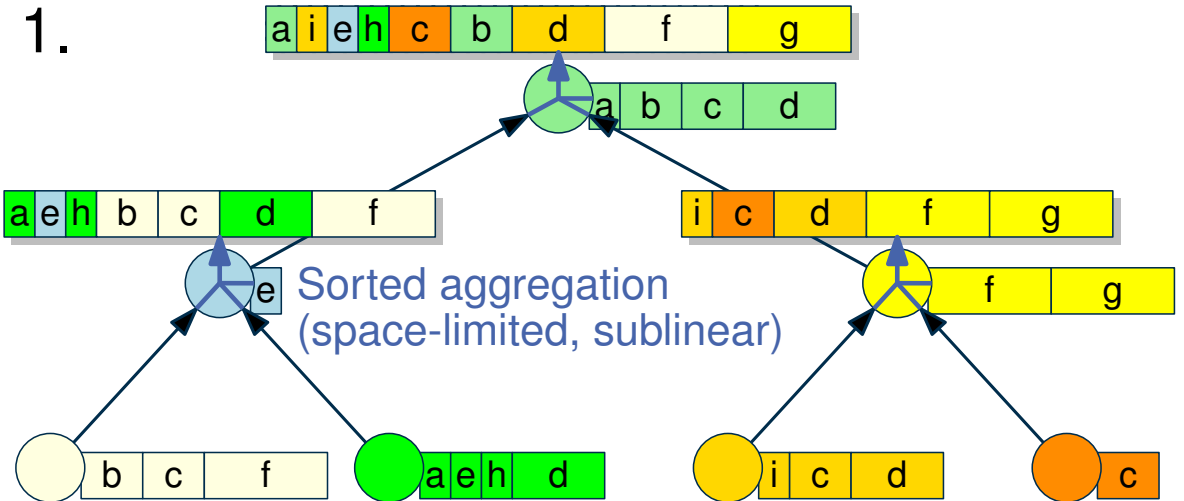
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Observations

- Clause needs to meet global quality threshold to be shared successfully
- Quality threshold adapts to state of solving



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The Problem

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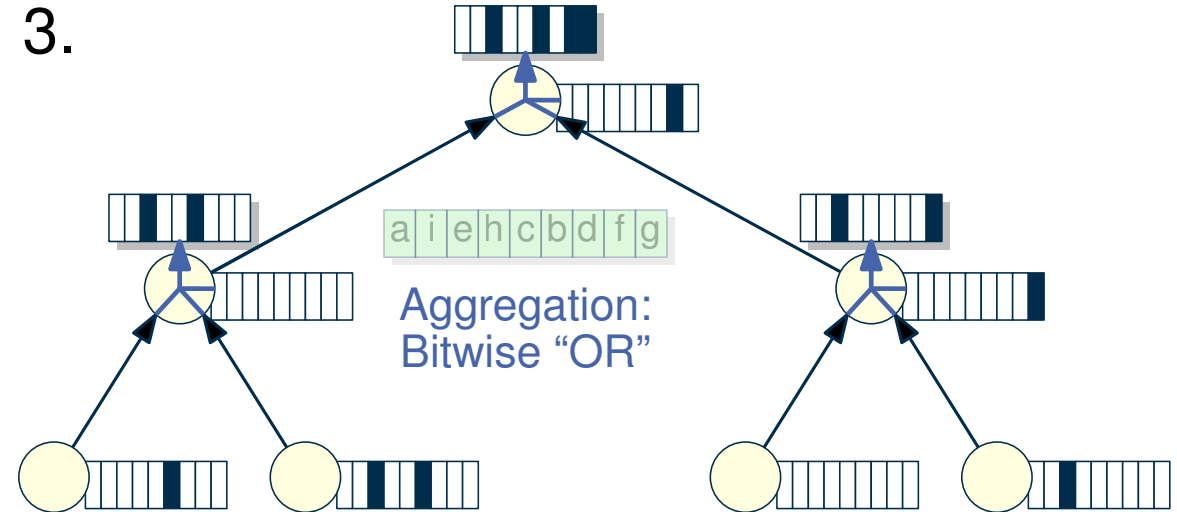
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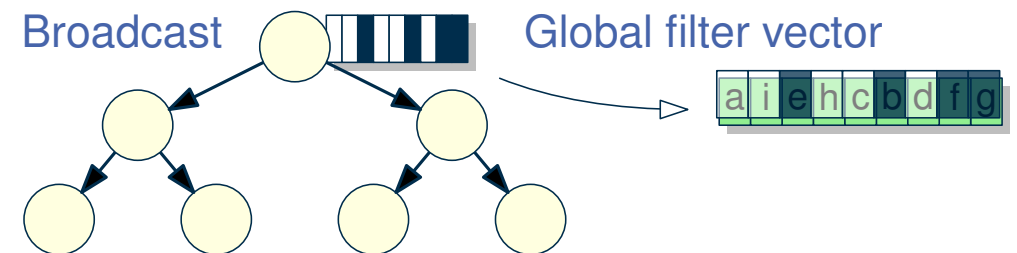
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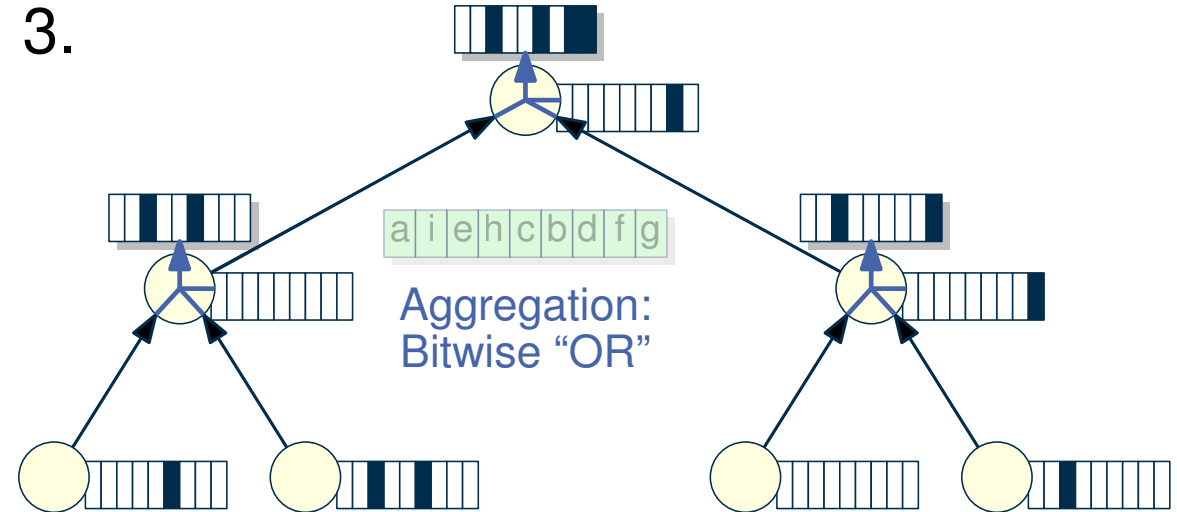
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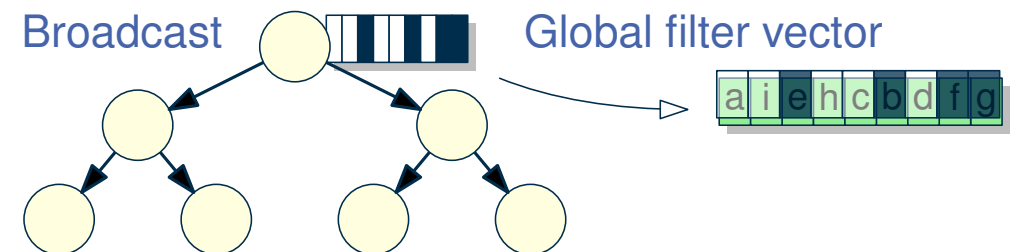
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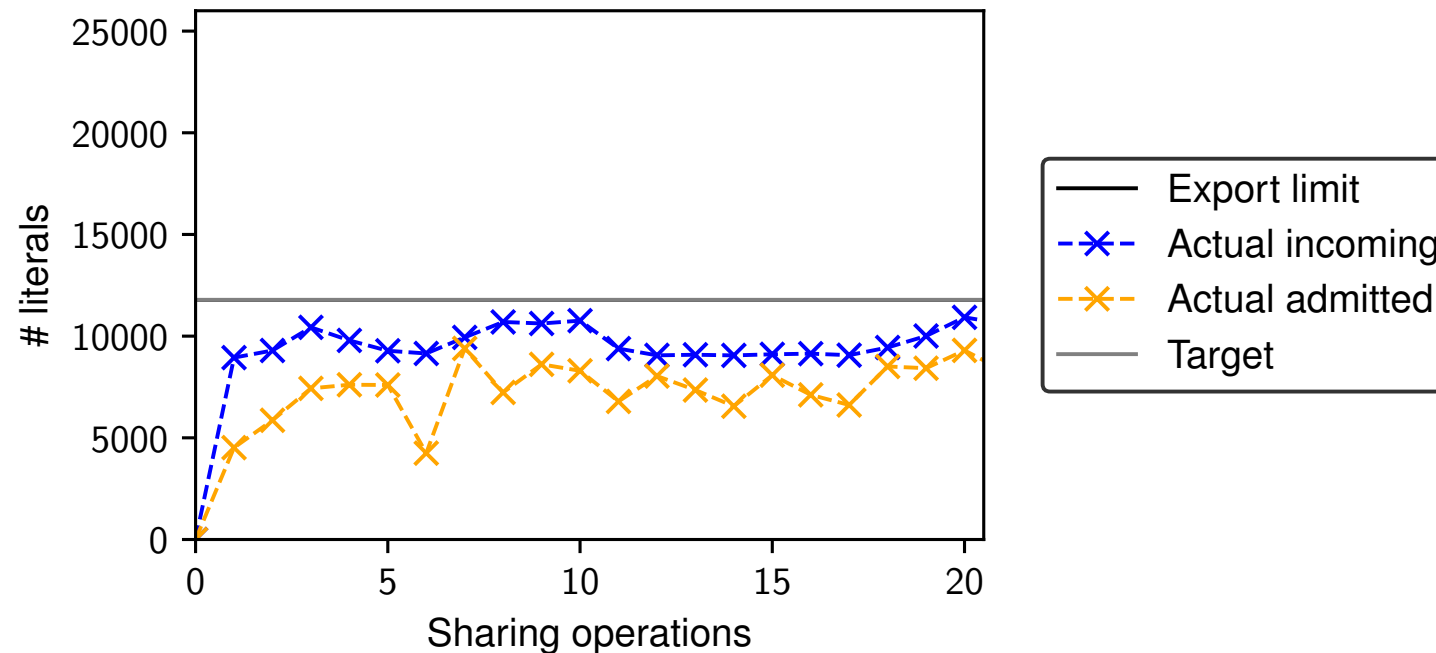
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We want to share L literals per sharing but may only get $L' < L$ successfully shared literals. Why?

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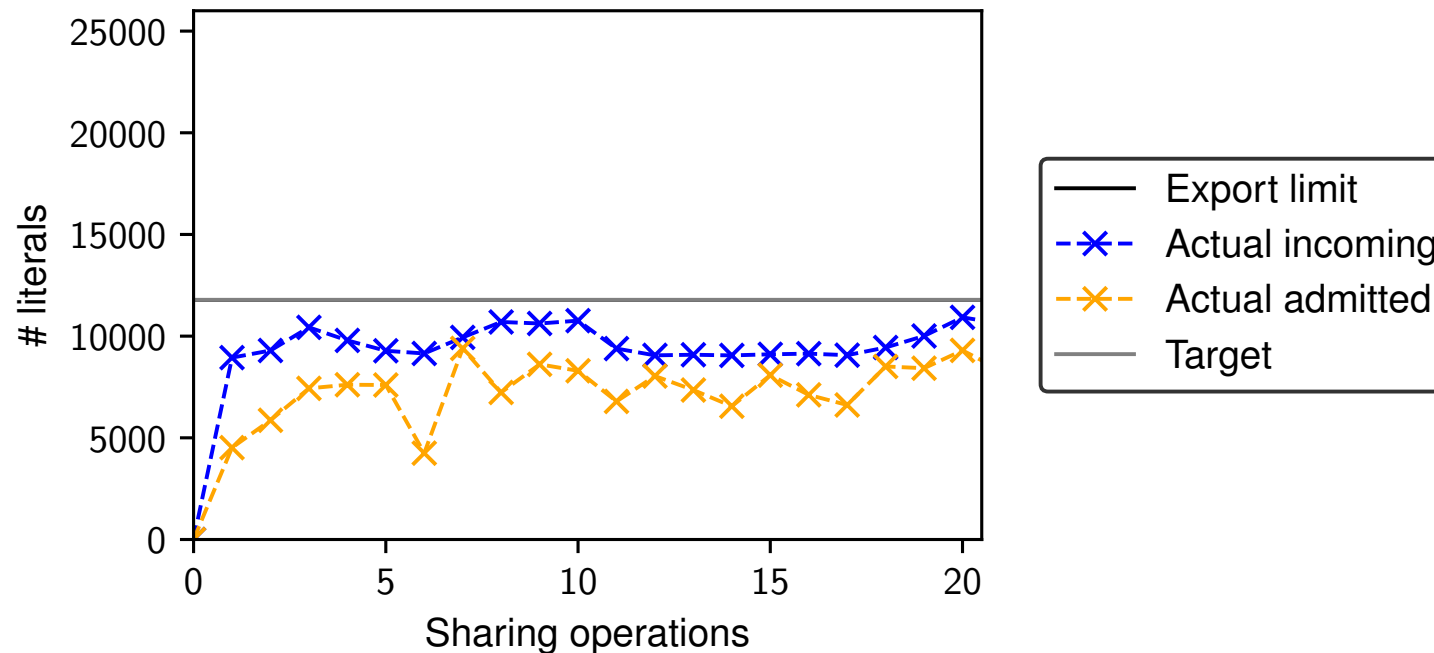


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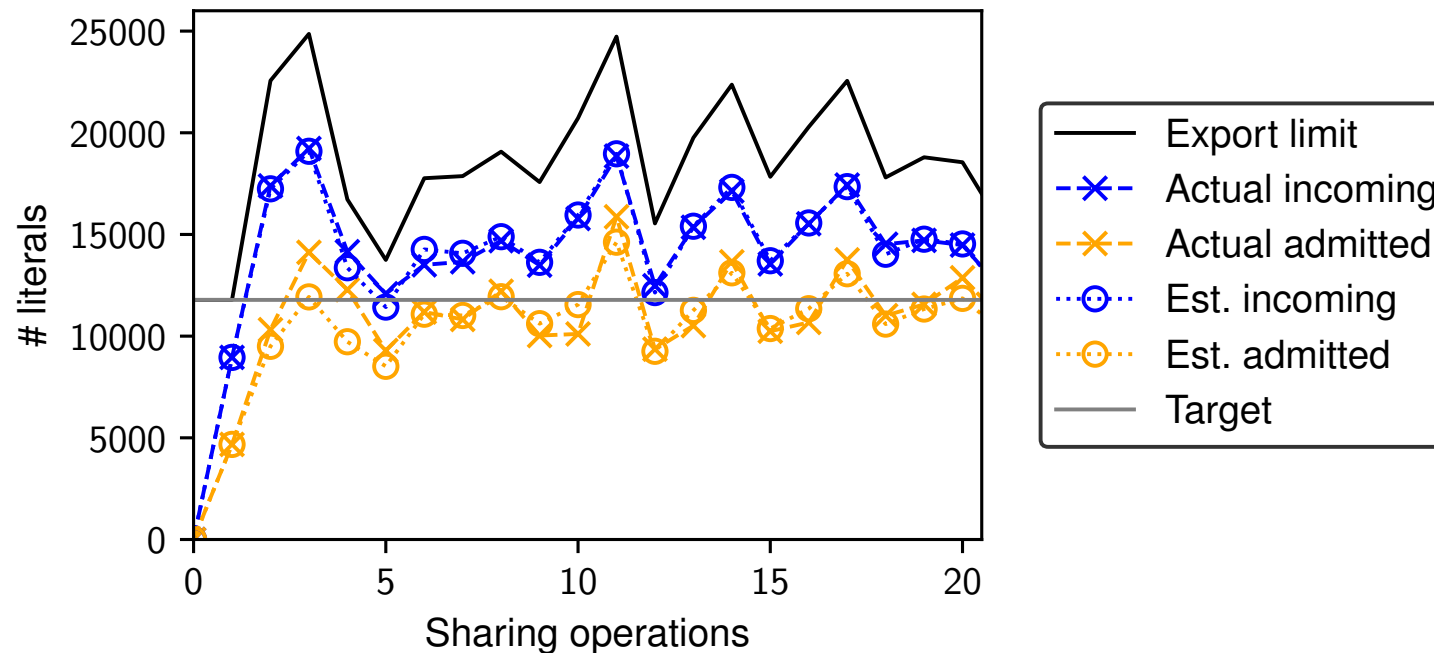


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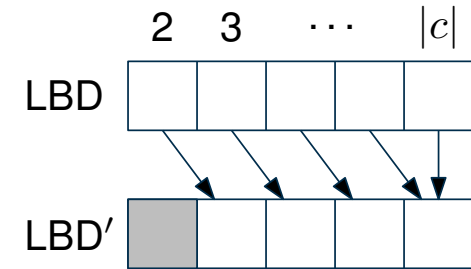
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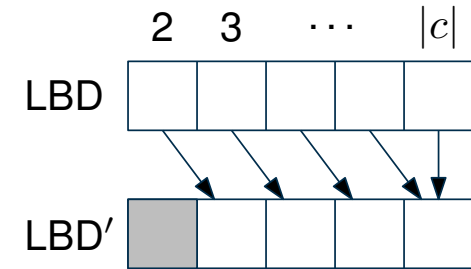
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But: Dropping LBD values altogether performs just as well in latest experiments [22]



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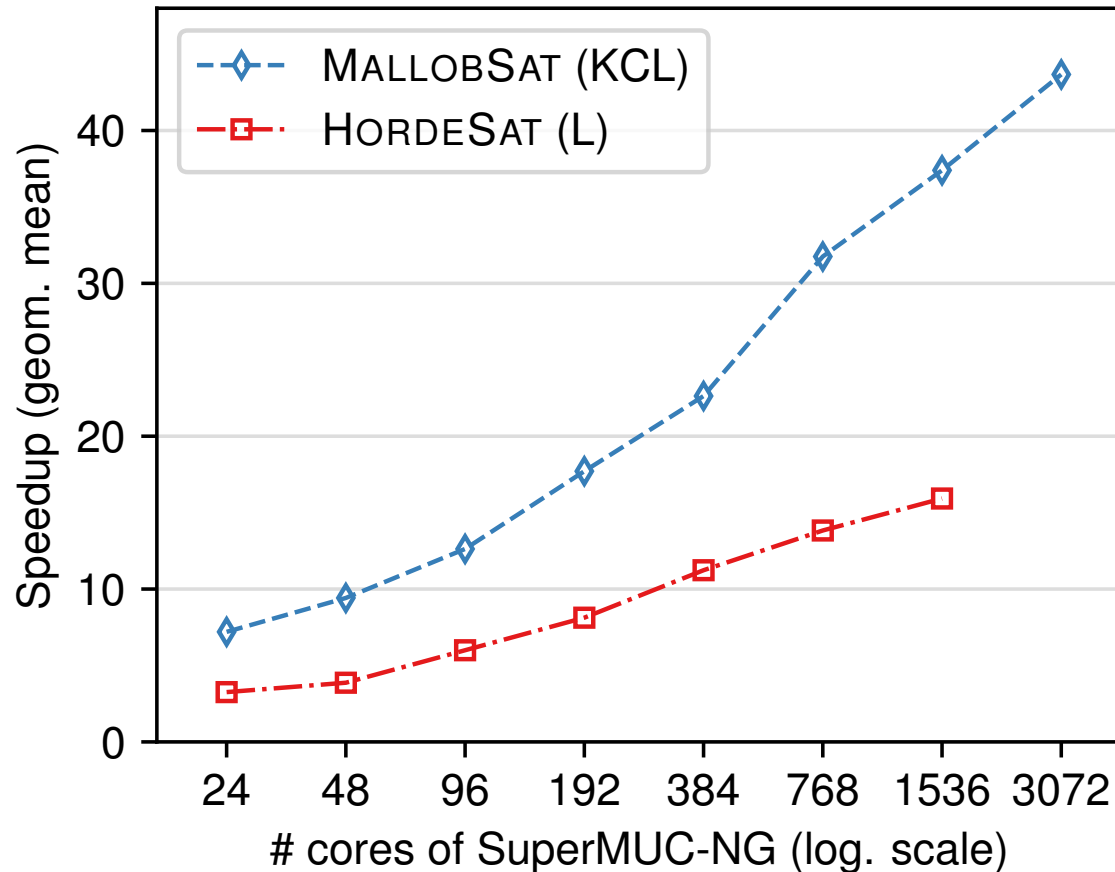
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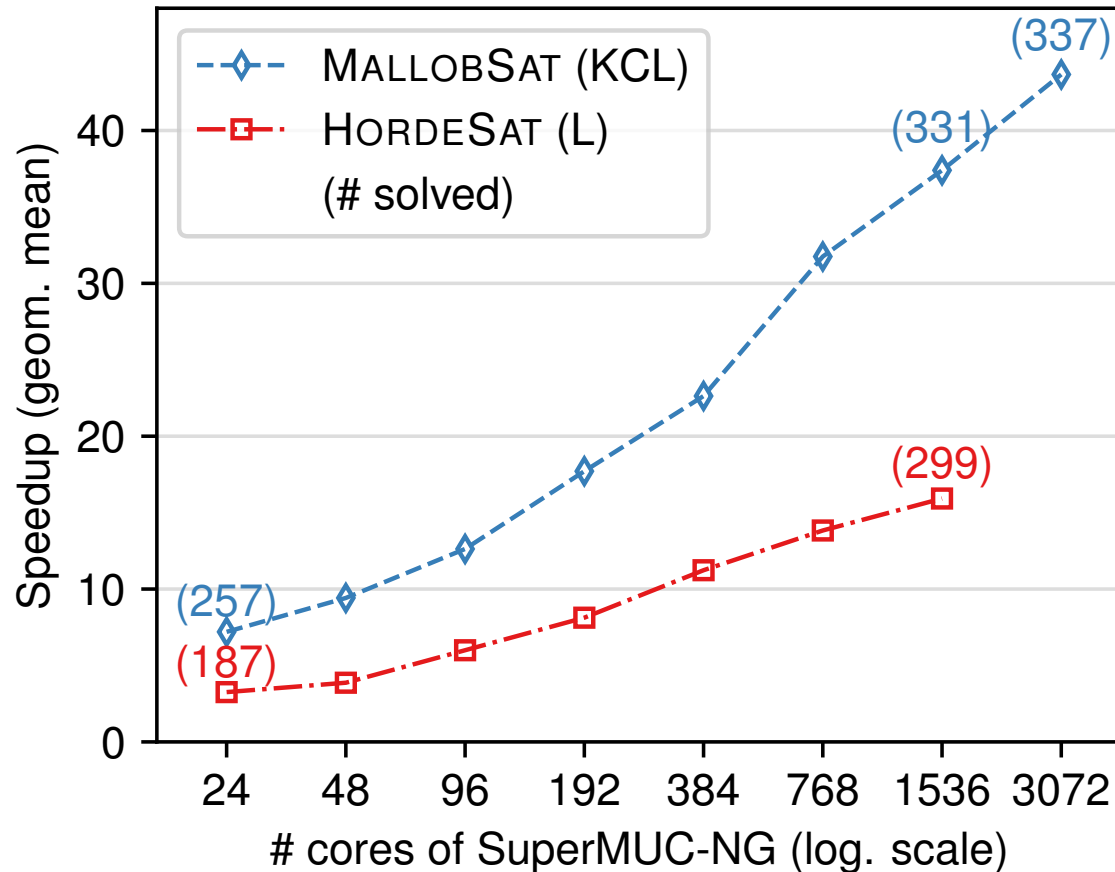
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Scaling of MallobSat [23]



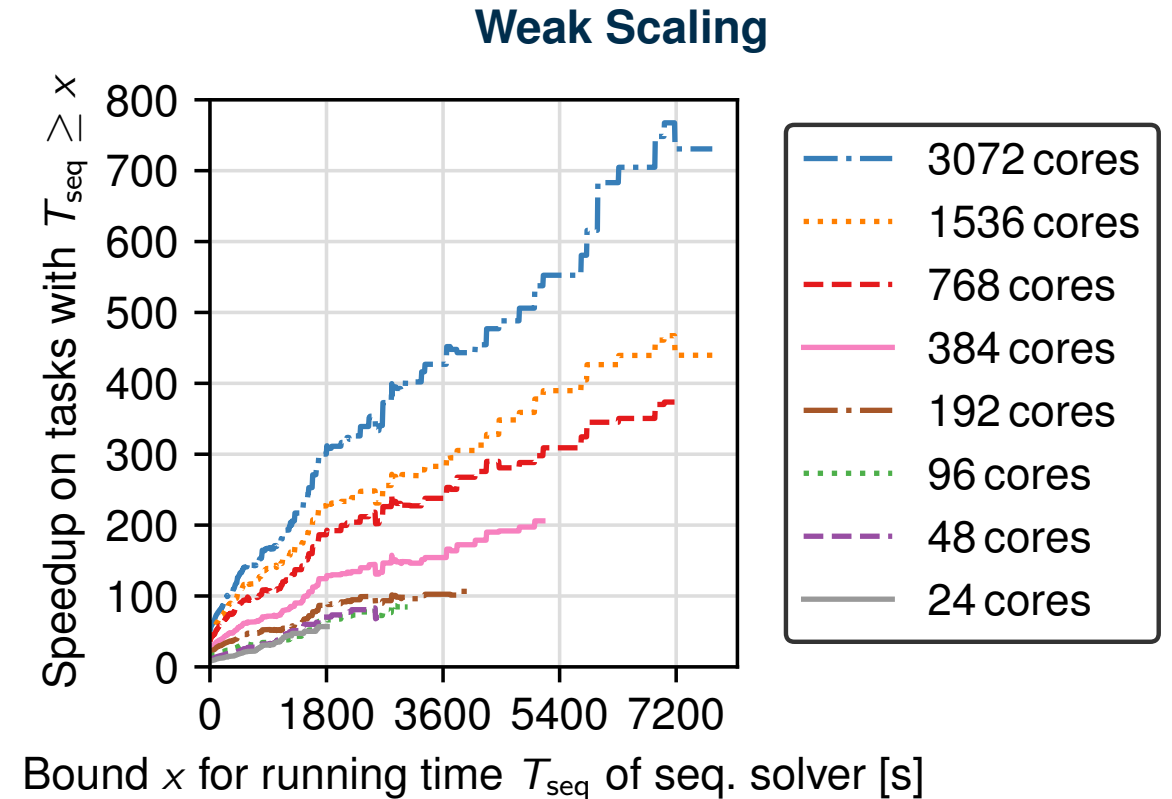
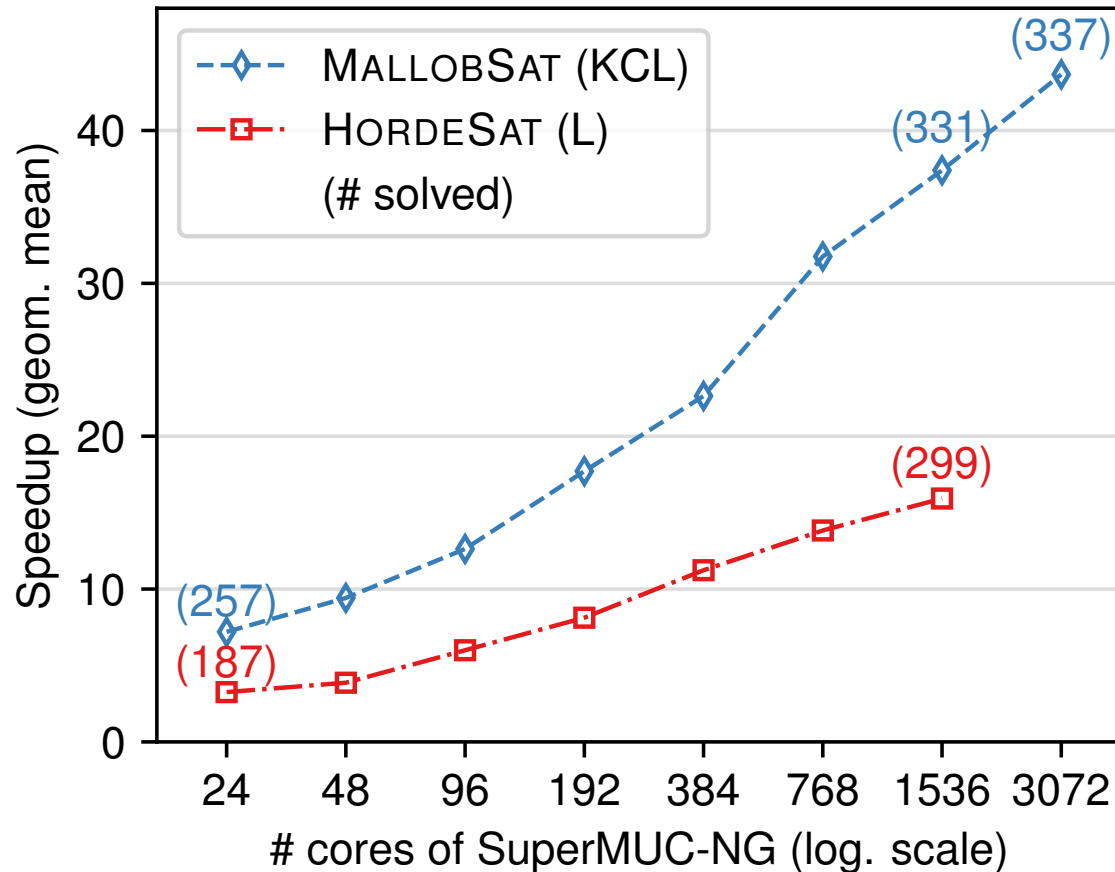
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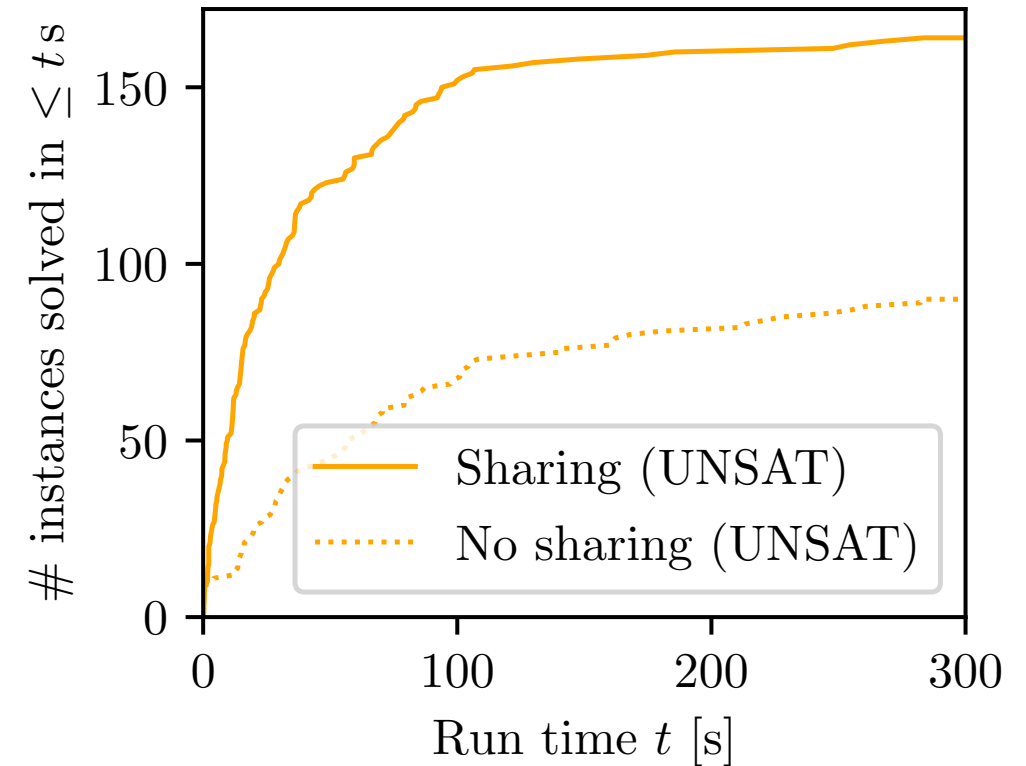
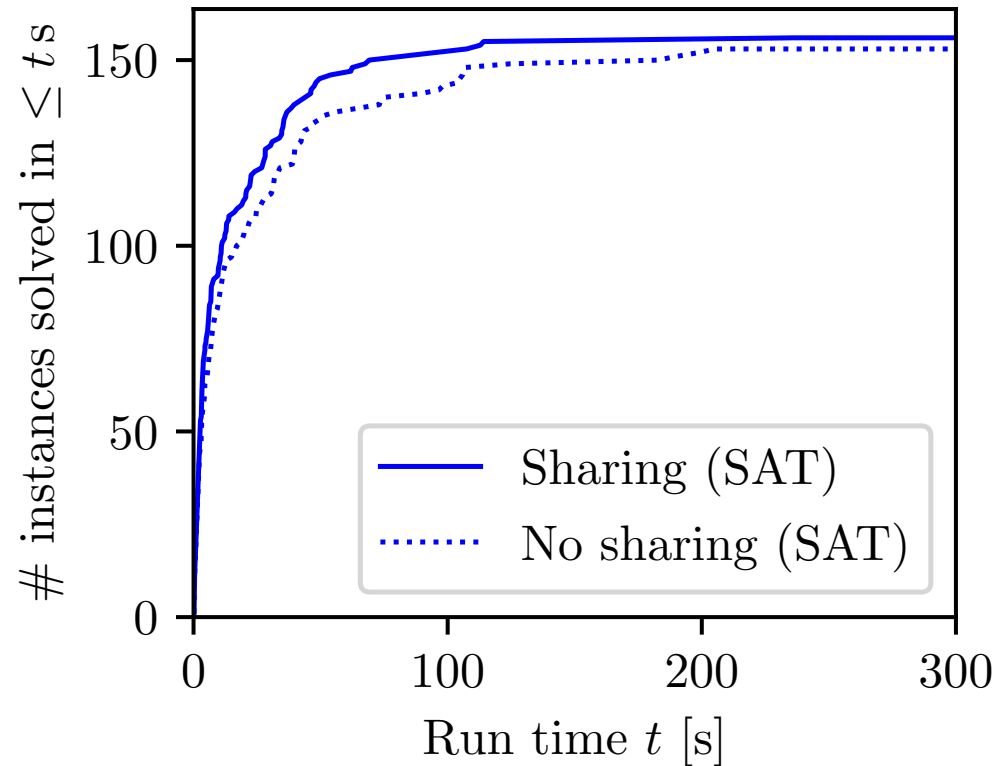
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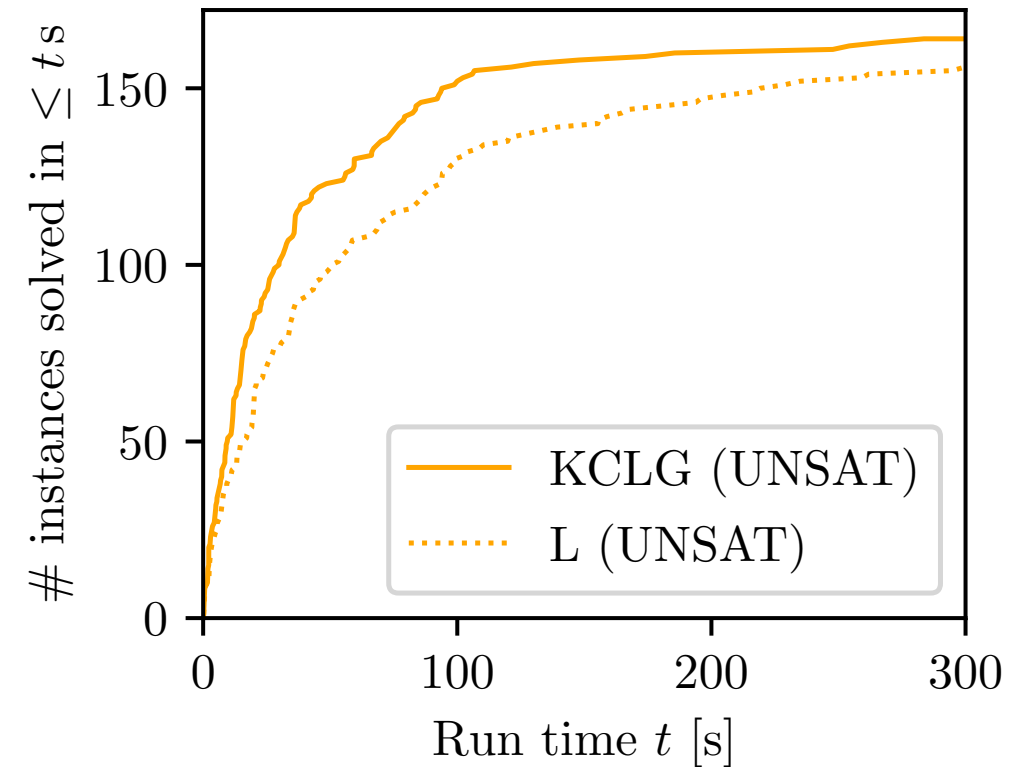
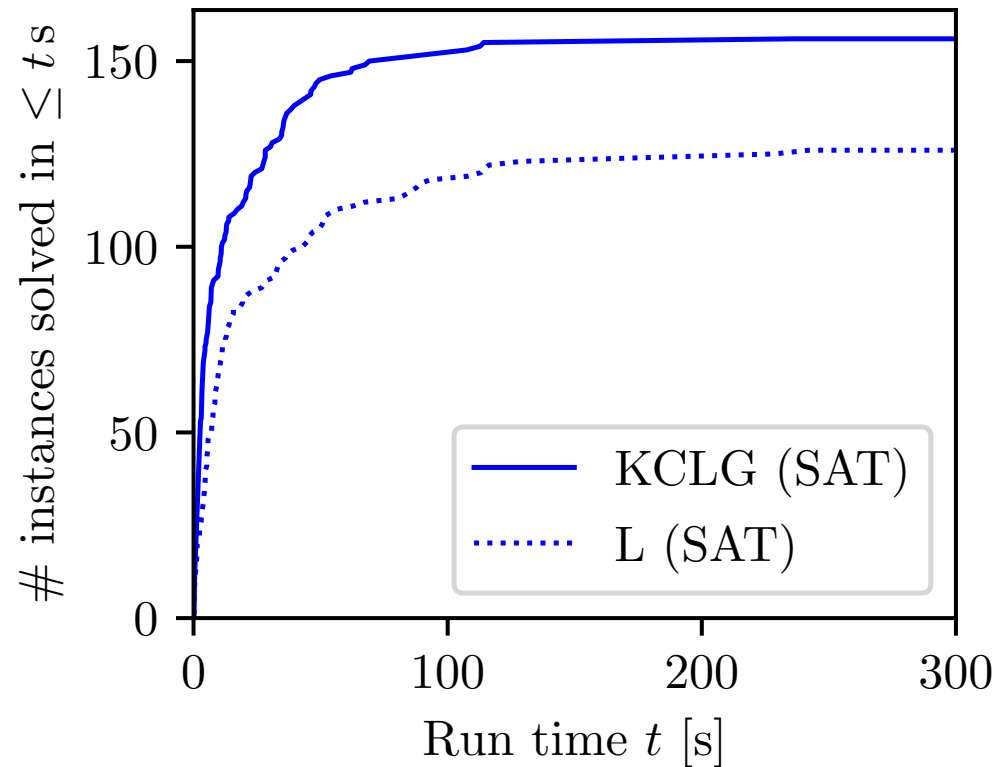
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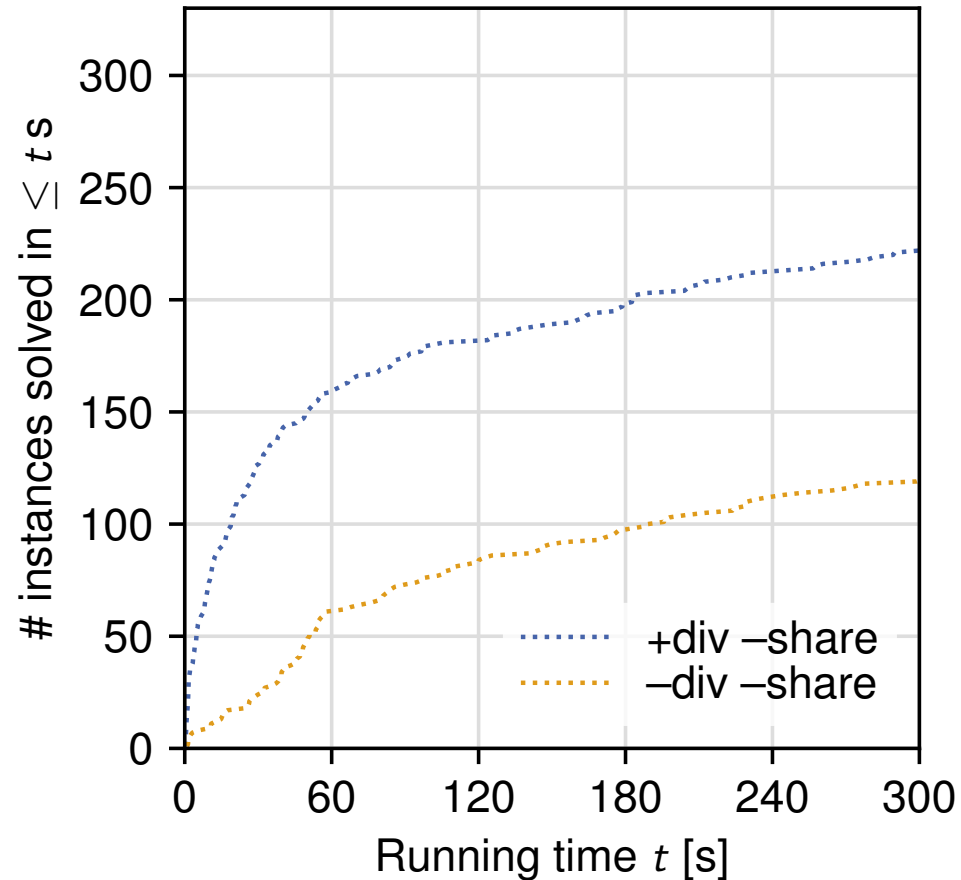
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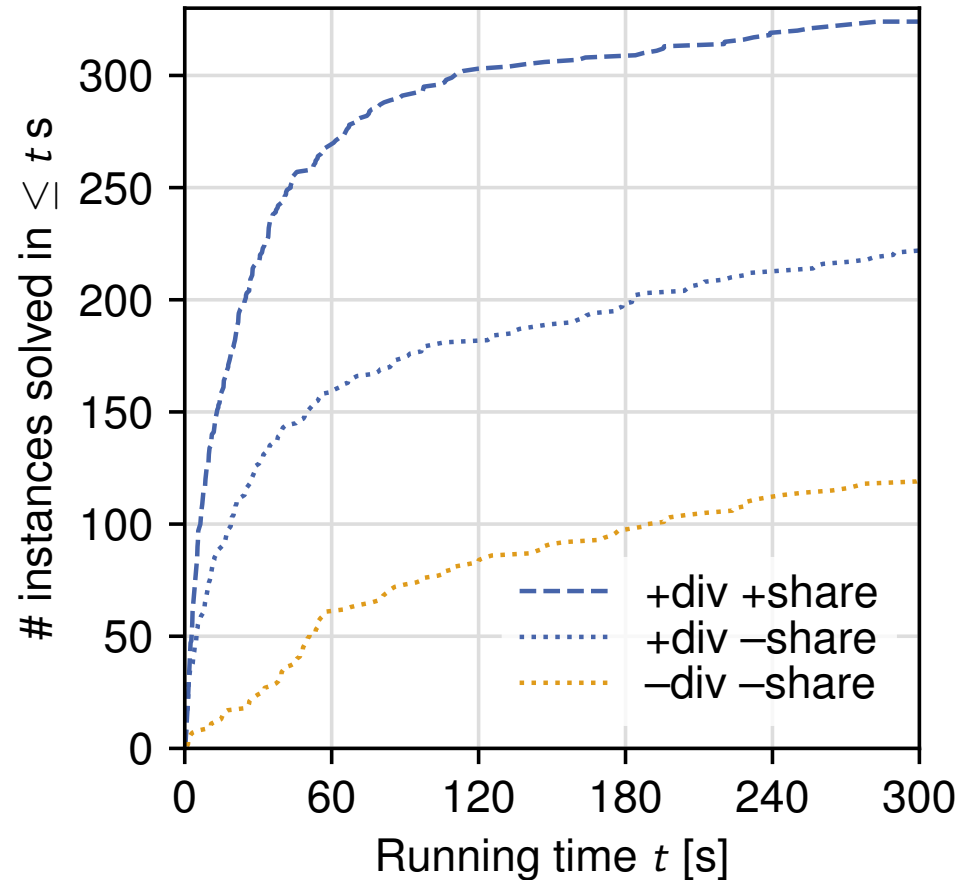
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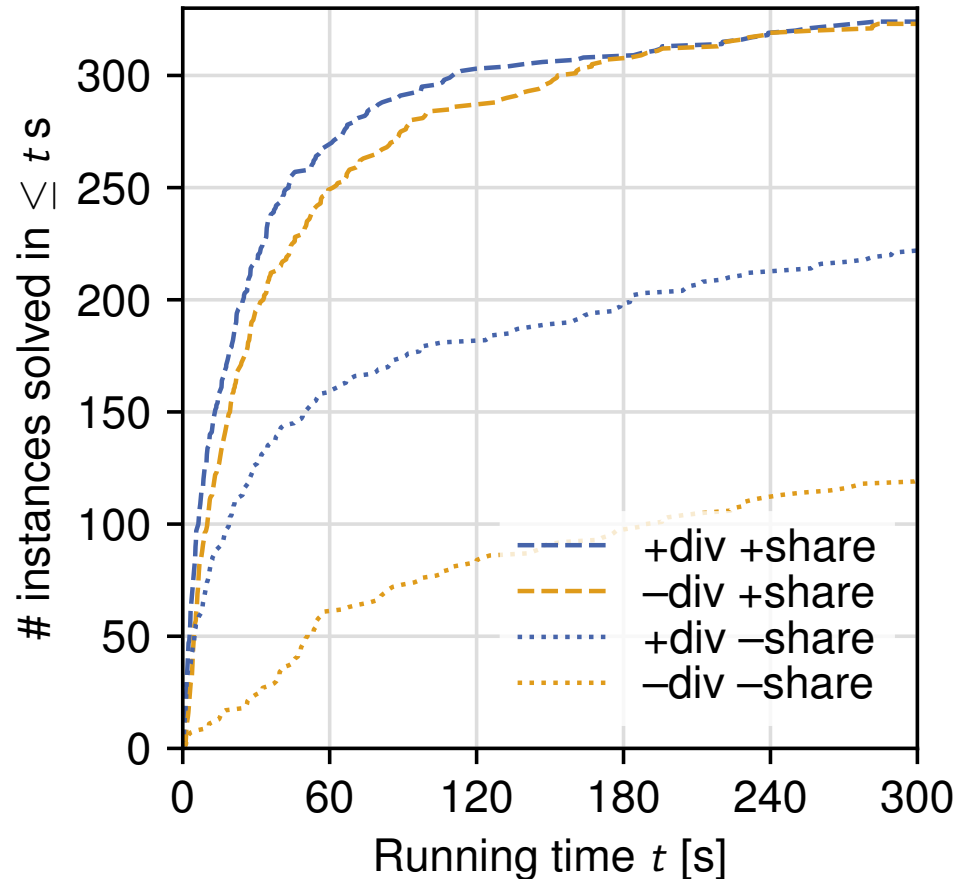
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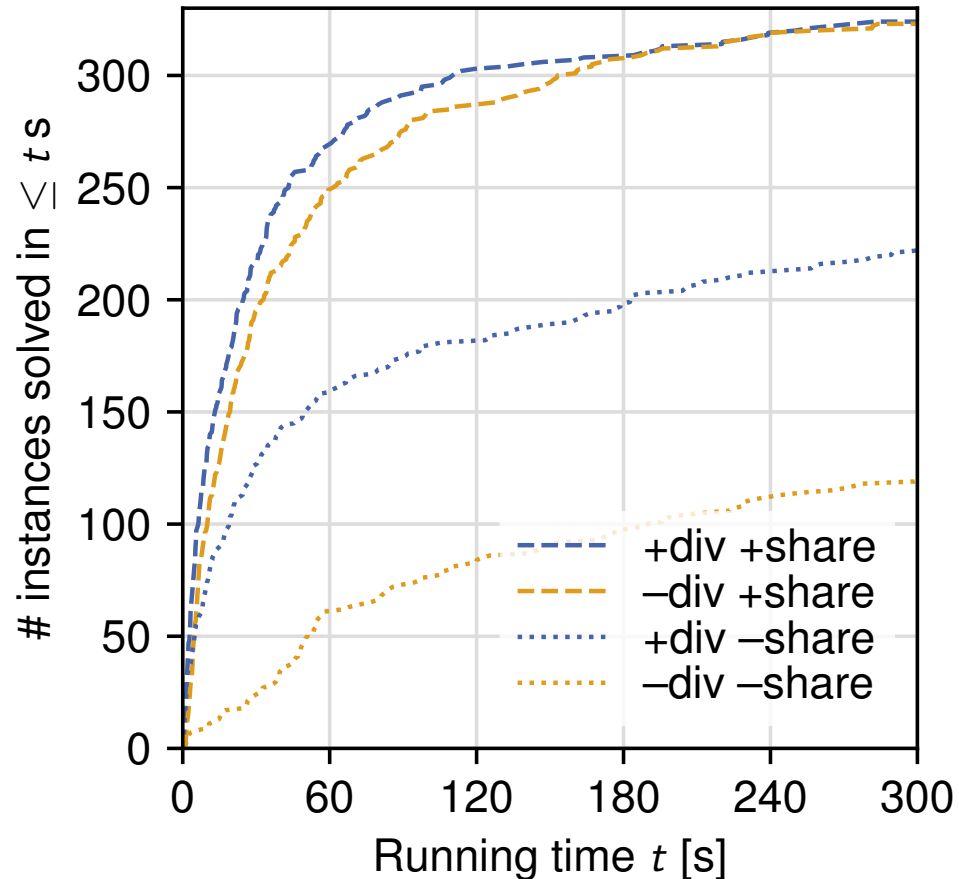
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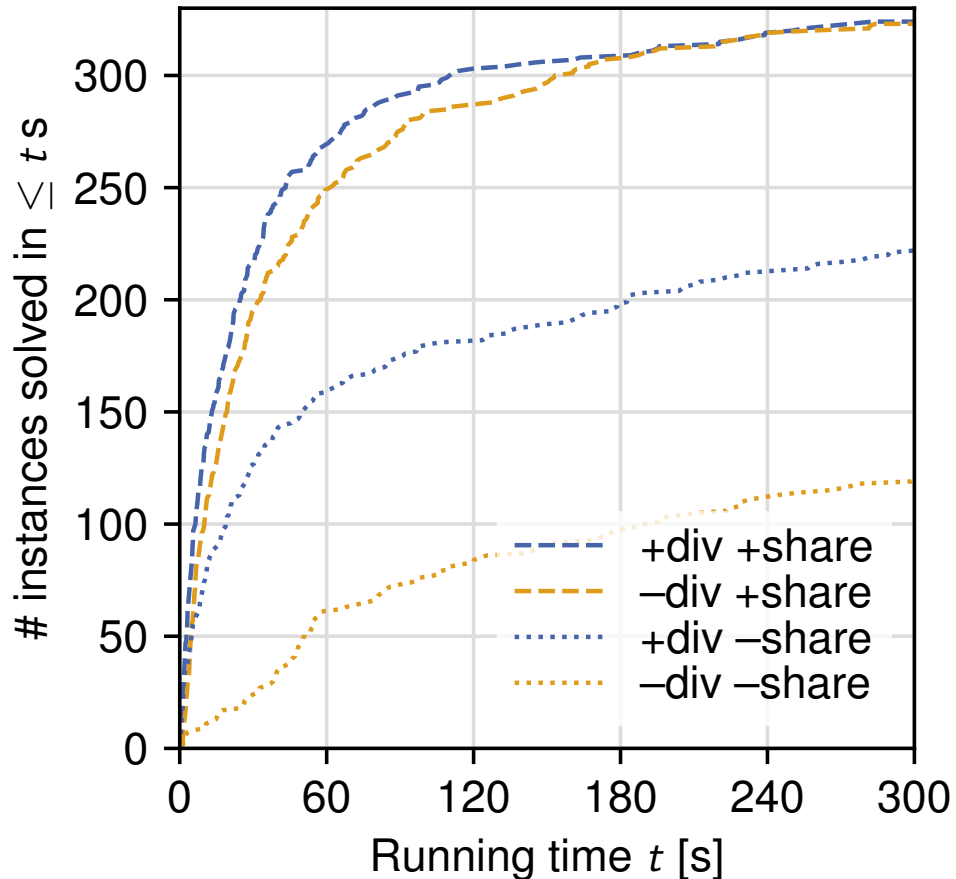
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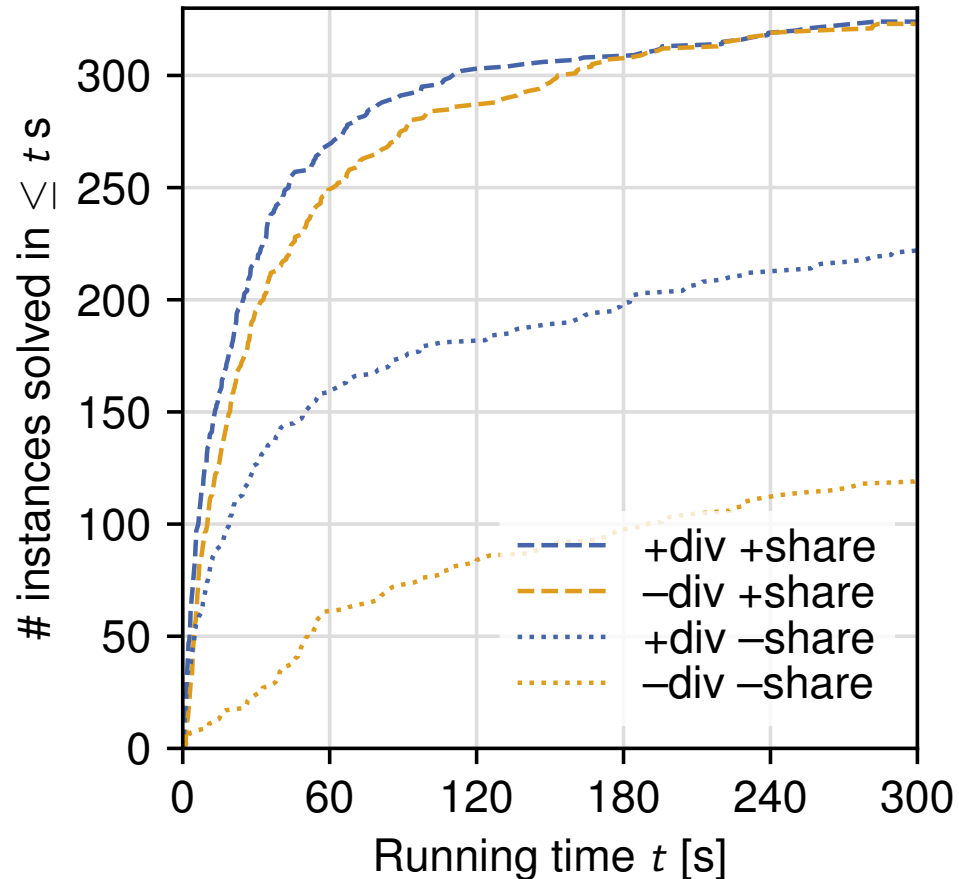
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- ⇒ Portfolio of diverse configurations is dispensable
- ⇒ Clause sharing is essential

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MallobSat: A Portfolio Solver?

Prevalent concept in literature: **Portfolio solver** with clause sharing / Clause-sharing portfolio

- “a problem instance is independently given to a collection of solvers *competing* for a solution in parallel”
– Fichte et al., 2023 [9]
- “each thread runs a *different SAT solver* on the same instance[, which] *in combination with clause-sharing* leads to surprisingly good performance *for small portfolio sizes*”
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Our view, based on empirical observations:

MALLOBSAT is a Clause-sharing solver with diversification

- Clause sharing = main driver of scalability
- Adding explicit diversification is beneficial but not essential
- Applicability to other solvers?

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Massive parallelism for a single formula

- Faster solving times
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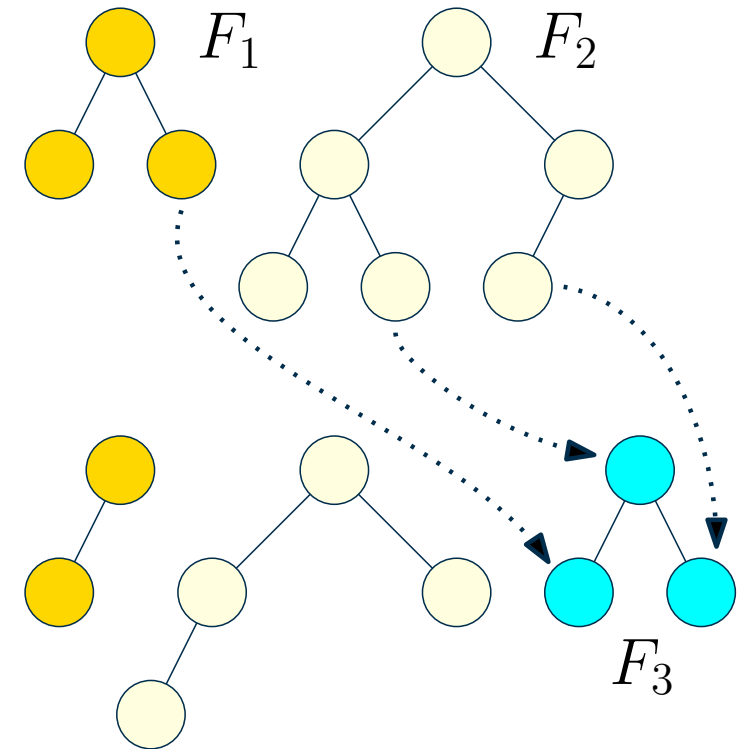
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Best of both worlds? [19]

- On demand scheduling of incoming (SAT) jobs
- Resize jobs during their execution as needed
- Few milliseconds to schedule an incoming job, full utilization whenever sufficient demand is present



Scheduling Many SAT Tasks [23]

Problem statement

Given $x \in \{400, 1600, 6400\}$ cores and 400 SAT tasks, solve as many tasks as possible.

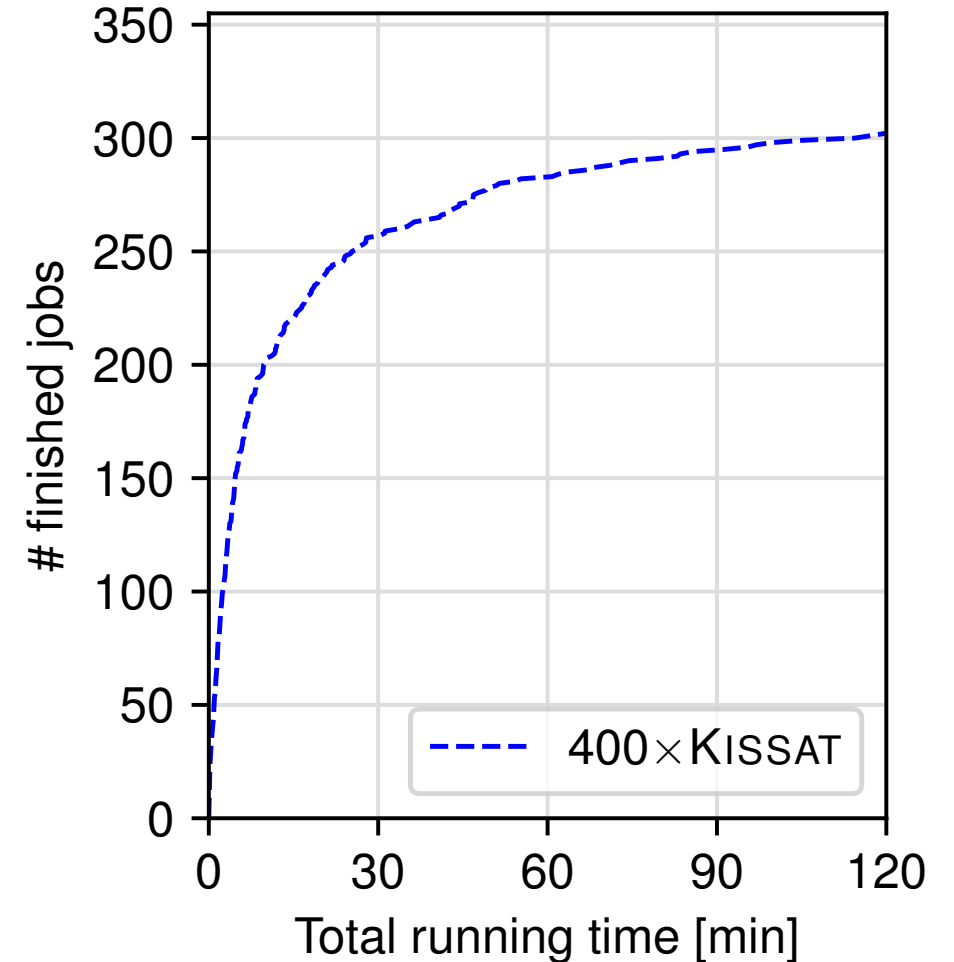
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- Highly efficient (no redundant work)
- Sequential solving only → no scaling opportunity



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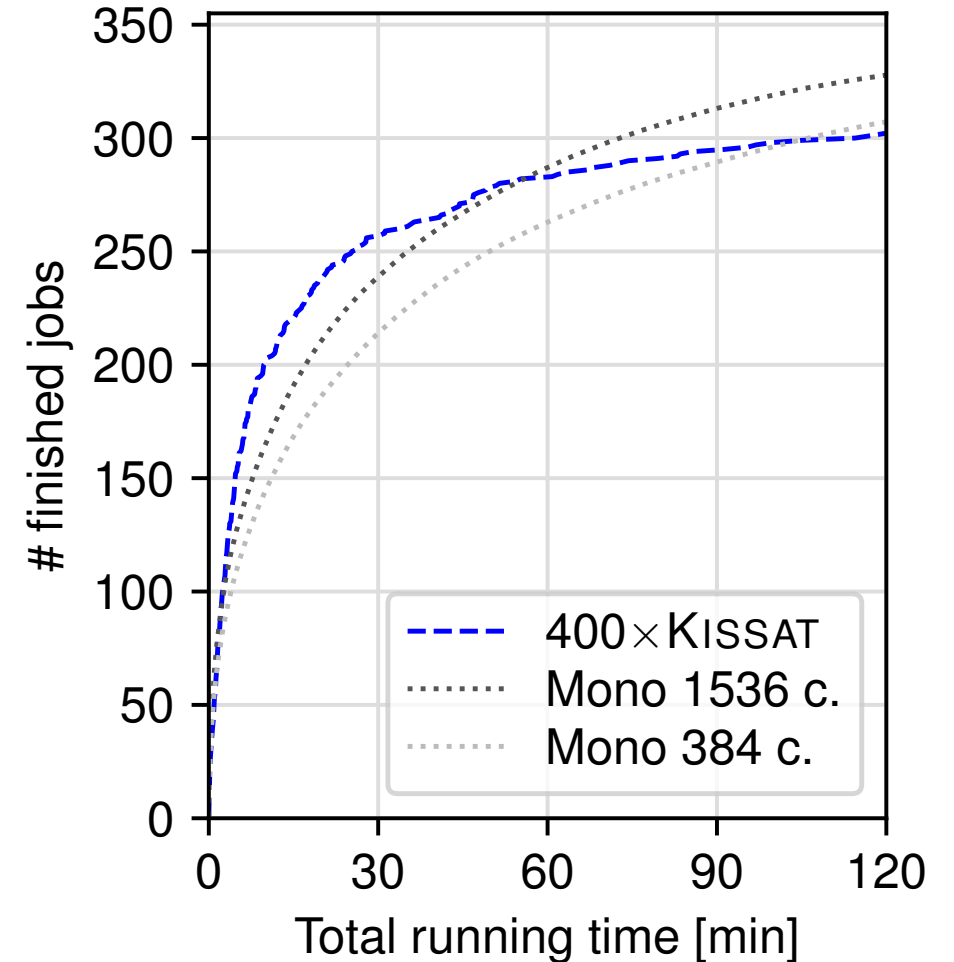
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Extreme 1: 400 sequential solvers ($400 \times$ Kissat)

Extreme 2: “Monolithic” parallel solving

- One job at a time
- Generous assumption: instances sorted by run time
- Maximizes per-instance speedups
- No task parallelism, poor efficiency



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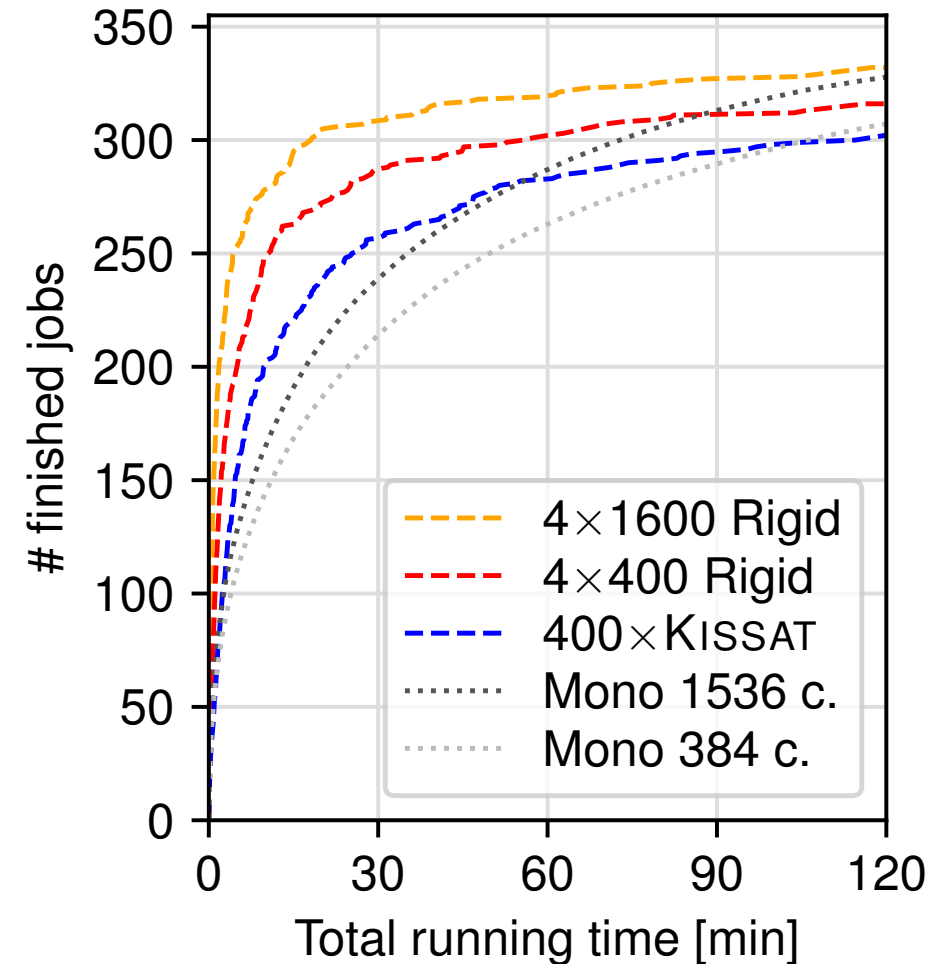
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Middle ground 1: Divide cores evenly among jobs

- Solid speedups at small-scale parallel SAT
- After 15 min, $> 50\%$ of cores are idling



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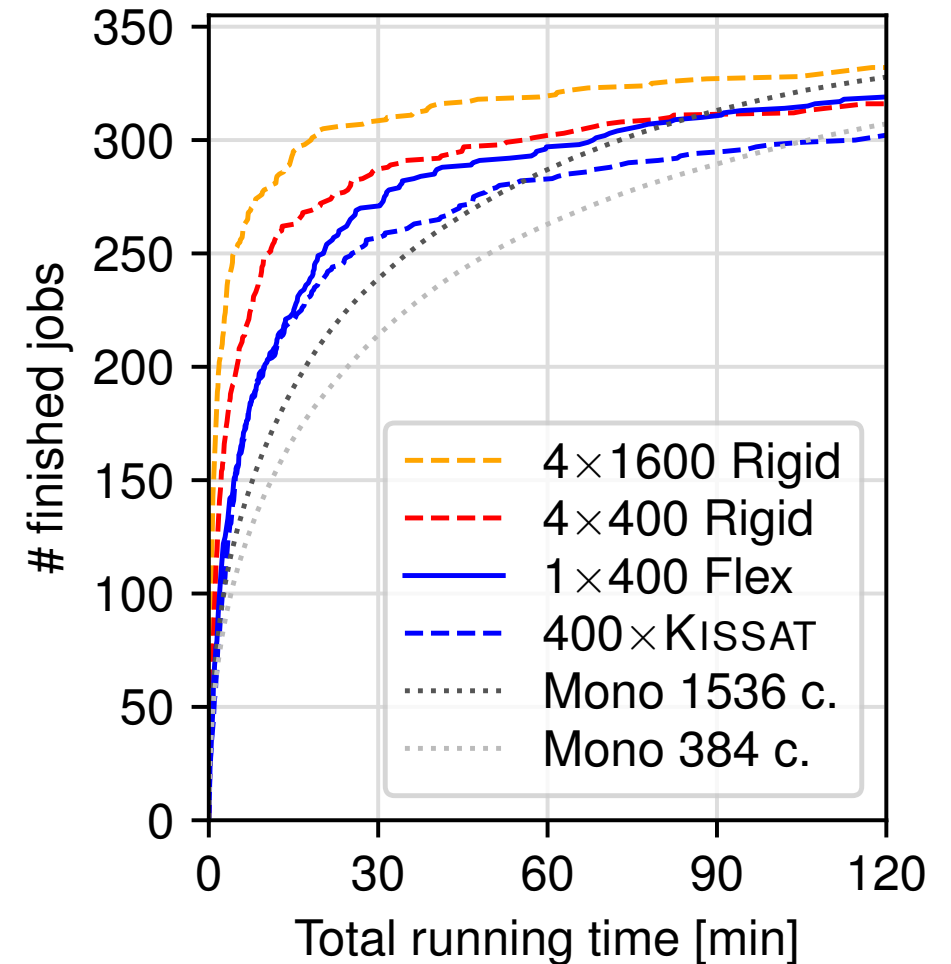
Extreme 1: 400 sequential solvers ($400 \times$ Kissat)

Extreme 2: “Monolithic” parallel solving

Middle ground 1: Divide cores among jobs (rigid)

Middle ground 2: Divide cores **dynamically** among jobs

- Finishing jobs **yield resources** to remaining jobs



Scheduling Many SAT Tasks [23]

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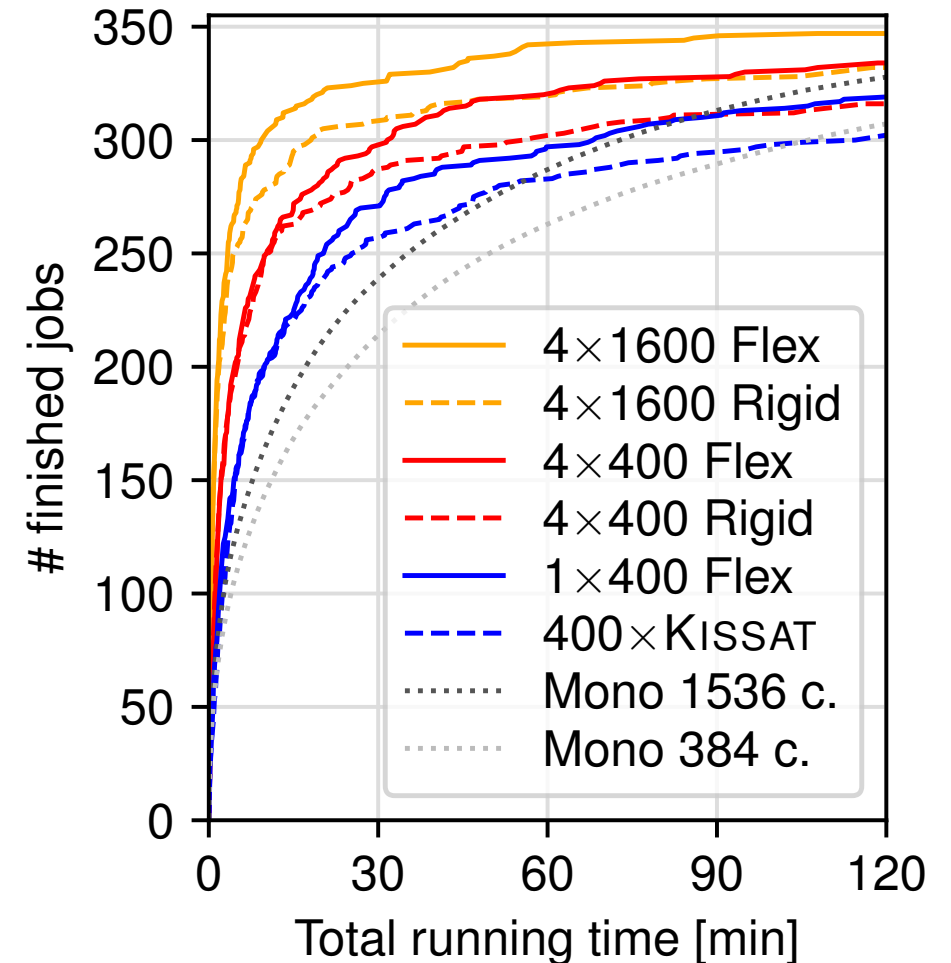
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- **Record number of solved instances** using **relatively little resources**



Current Frontiers

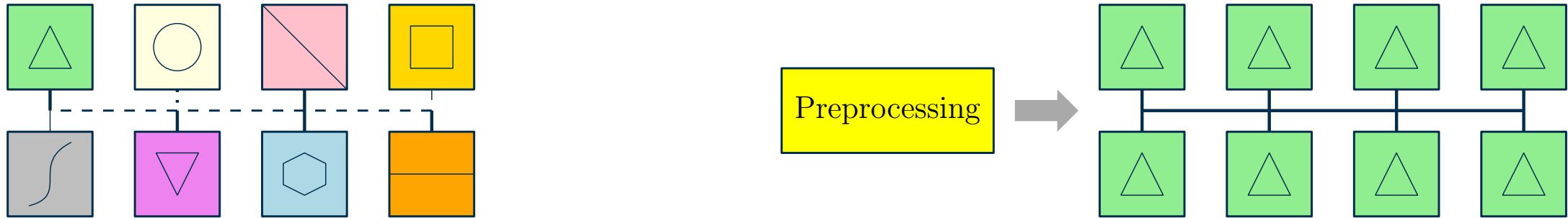
- Streamlined clause-sharing solver design, away from portfolios [22]



Clause sharing portfolio with “language barriers” **vs.** streamlined parallel CDCL

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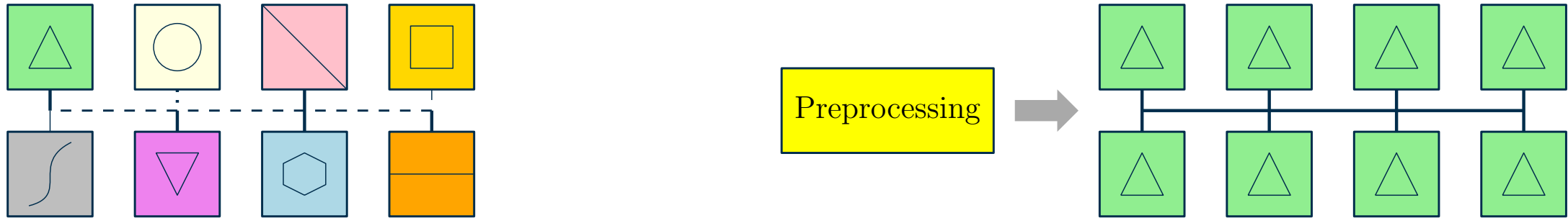


Clause sharing portfolio with “language barriers” **vs.** streamlined parallel CDCL

- Address **high main memory requirements**, especially for huge formulas (cf. [10])
- Preprocessing and Inprocessing in parallel SAT solving (cf. [16])

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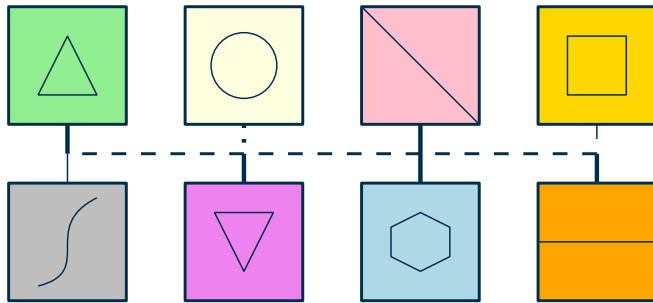


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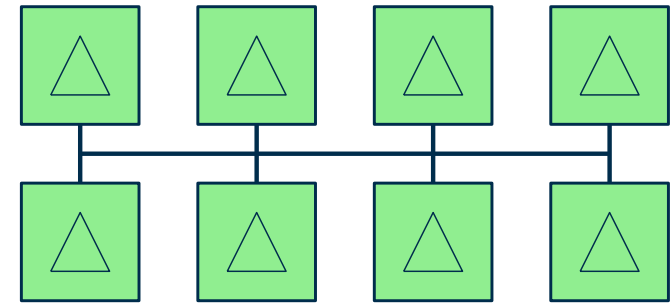
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- Applying technology to related tools, applications
 - Verification (Bounded Model Checking, SMT Solving)
 - Optimization (MaxSAT Solving)
 - Crucial piece of technology: Distributed **incremental SAT solving!** [21]

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Preprocessing

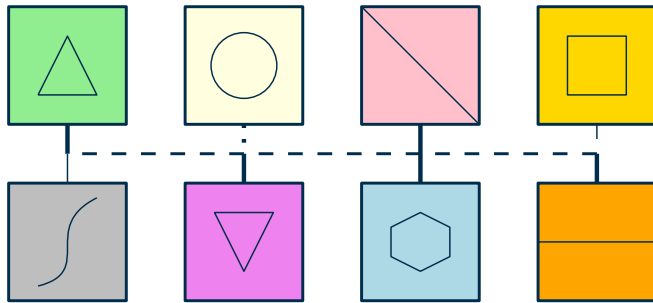


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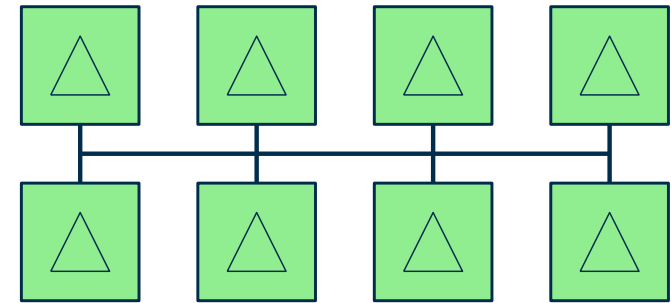
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- **Uses of GPUs ??** (Stochastic Local Search [6], Inprocessing offloading [16], ...)
- Proofs of unsatisfiability (next lecture!)

Recap

Parallel SAT Solving

- Popular parallelization approaches for SAT (“antisocial nerds” analogy)
 - Search space splitting, Cube&Conquer
 - Pure portfolio
 - Clause sharing portfolio
- All-to-all clause sharing can be very useful and scalable (up and down) **if implemented well**
 - diversifies solvers effectively in and of itself
- Exploit embarrassingly parallel job processing for interactive solving & best efficiency

Next Up

- Proof pragmatics: Formats for different proof systems in the wild
- Bringing proof technology to parallel and distributed SAT solving

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